

Treatment/Management Questions

Category: Treatment/Management **Total Questions:** 206 **Purpose:** Medical professional review of categorization accuracy

Review Instructions

For each question below, please mark:

- ✓ **Correctly categorized** - Question clearly belongs in this category
- ✗ **Miscategorized** - Question belongs in a different category (note which one)
- ? **Ambiguous** - Question could reasonably fit multiple categories

You don't need to review all questions! Even 20-30 per category is very helpful.

What 'Treatment/Management' Means

Questions asking which drug to prescribe, which therapy to choose, or how to manage the condition.

Questions for Review

Question 1

Clinical Scenario:

A 62-year-old man comes to the physician for a follow-up examination. One month ago, therapy with lisinopril was initiated for treatment of hypertension. His blood pressure is 136/86 mm Hg. Urinalysis shows a creatinine clearance of 92 mL/min. The patient's serum creatinine concentration is most likely closest to which of the following values?

Answer Choices:

A. 2.3 mg/dL B. 2.0 mg/dL C. 1.1 mg/dL D. 1.7 mg/dL

Correct Answer: C. 1.1 mg/dL

Question 2

Clinical Scenario:

A 57-year-old man presents to his oncologist to discuss management of small cell lung cancer. The patient is a lifelong smoker and was diagnosed with cancer 1 week ago. The patient states that the cancer was his fault for smoking and that there is "no hope now." He seems disinterested in discussing the treatment options and making a plan for treatment and followup. The patient says "he does not want any treatment" for his condition. Which of the following is the most appropriate response from the physician?

Answer Choices:

A. "I respect your decision and we will not administer any treatment. Let me know if I can help in any way." B. "It must be very challenging having received this diagnosis. I want to work with you to create a plan." C. "We are going to need to treat your lung cancer. I am here to help you throughout the process." D. "You seem upset at the news of this diagnosis. I want you to go home and discuss this with your loved ones and come back when you feel ready to make a plan together for your care."

Correct Answer: B. "It must be very challenging having received this diagnosis. I want to work with you to create a plan."

Question 3

Clinical Scenario:

A 65-year-old man with hypertension comes to the physician for a routine health maintenance examination. Current medications include atenolol, lisinopril, and atorvastatin. His pulse is 86/min, respirations are 18/min, and blood pressure is 145/95 mm Hg. Cardiac examination is shown. Which of the following is the most likely cause of this physical examination finding?

Answer Choices:

A. Decreased compliance of the left ventricle B. Myxomatous degeneration of the mitral valve C. Inflammation of the pericardium D. Dilation of the aortic root

Correct Answer: A. Decreased compliance of the left ventricle

Question 4

Clinical Scenario:

A 17-year-old male comes to the physician because of painful genital sores, malaise, and fever for 3 days. He is sexually active with 3 female partners and does not use condoms consistently. His temperature is 38.3°C (101°F). Physical examination shows tender lymphadenopathy in the left inguinal region and multiple, punched-out ulcers over the penile shaft and glans. Microscopic examination of a smear from the ulcer is most likely to show which of the following?

Answer Choices:

A. Eosinophilic intranuclear inclusions B. Basophilic intracytoplasmic inclusions C. Eosinophilic intracytoplasmic inclusions D. Basophilic intranuclear inclusions

Correct Answer: A. Eosinophilic intranuclear inclusions

Question 5

Clinical Scenario:

A 22-year-old man, accompanied by his brother, presents to the emergency department with palpitations for the past 30 minutes and nausea for the past hour. When the patient meets the physician, he says, "Doctor, I am the happiest person in the world because I have the best brain possible. It's just that my heart is saying something, so I came to check with you to see what it is". The brother says the patient was diagnosed with attention-deficit/hyperactivity disorder (ADHD) 5 years ago. When the doctor asks the patient about his ADHD treatment, he replies, "Doctor, the medicine is wonderful, and I love it very much. I often take one or two tablets extra!" He has no history of a known cardiovascular disorder, alcohol abuse, or smoking. The patient's temperature is 99.2°F (37.3°C), heart rate is 116/minute, respiratory rate is 18/minute, and blood pressure is 138/94 mm Hg. Generalized perspiration is present. Which of the following signs is most likely to be present on ocular examination?

Answer Choices:

A. Dilated pupils B. Rotatory nystagmus C. Bilateral foveal yellow spots D. Bilateral optic disc edema

Correct Answer: A. Dilated pupils

Question 6

Clinical Scenario:

A 42-year-old woman comes to the physician because of a 2-week history of joint pain and fatigue. She has a history of multiple unprovoked deep vein thromboses. Physical examination shows small bilateral knee effusions and erythematous raised patches with scaling and follicular plugging over the ears and scalp. Oral examination shows several small ulcers. Laboratory evaluation of this patient is most likely to show which of the following?

Answer Choices:

A. Positive rapid plasma reagin test B. Positive anti-citrullinated peptide antibodies C. Decreased activated partial thromboplastin time D. Negative antinuclear antibodies

Correct Answer: A. Positive rapid plasma reagin test

Question 7

Clinical Scenario:

A 78-year-old woman is brought to the emergency ward by her son for lethargy and generalized weakness. The patient speaks in short utterances and does not make eye contact with the provider or her son throughout the interview and examination. You elicit that the patient lives with her son and daughter-in-law, and she reports vague weakness for the last couple days. The emergency room provider notices 3-4 healing bruises on the patient's upper extremities; otherwise, examination is not revealing. Routine chemistries and blood counts are unremarkable; non-contrast head CT demonstrates normal age-related changes. Which of the following is the most appropriate next step in management?

Answer Choices:

A. Perform lumbar puncture B. Question the patient's son regarding the home situation C. Ask the patient's son to leave the room D. Call Adult Protective Services to report the patient's son

Correct Answer: C. Ask the patient's son to leave the room

Question 8

Clinical Scenario:

A 42-year-old woman is brought to the emergency department because of intermittent sharp right upper quadrant abdominal pain and nausea for the past 10 hours. She vomited three times. There is no associated fever, chills, diarrhea, or urinary symptoms. She has two children who both attend high school. She appears uncomfortable. She is 165 cm (5 ft 5 in) tall and weighs 86 kg (190 lb); BMI is 32 kg/m². Her temperature is 37°C (98.6°F), pulse is 100/min, and blood pressure is 140/90 mm Hg. She has mild scleral icterus. The abdomen is soft and nondistended, with tenderness to palpation of the right upper quadrant without guarding or rebound. Bowel sounds are normal. Laboratory studies show: Hemoglobin count 14 g/dL Leukocyte count 9,000/mm³ Platelet count 160,000/mm³ Serum Alkaline phosphatase 238 U/L Aspartate aminotransferase 60 U/L Bilirubin Total 2.8 mg/dL Direct 2.1 mg/dL Which of the following is the most appropriate next step in diagnosis?"

Answer Choices:

A. CT scan of the abdomen B. Transabdominal ultrasonography C. Endoscopic retrograde cholangiopancreatography D. HIDA scan of the biliary tract

Correct Answer: B. Transabdominal ultrasonography

Question 9

Clinical Scenario:

A 24-year-old woman comes to the physician because of a 3-day history of lower abdominal pain and dysuria. She has a history of recurring urinary tract infections that have resolved with antibiotic treatment. She is sexually active with one male partner and they do not use condoms. She had mild pain during her last sexual intercourse one week ago. Her temperature is 38.2°C (100.8°F), pulse is 86/min, and blood pressure is 110/70 mm Hg. Physical examination shows lower abdominal tenderness and bilateral inguinal lymphadenopathy. There is a small amount of purulent vaginal discharge. Bimanual examination shows uterine and cervical

motion tenderness. Laboratory studies show: Hemoglobin 12 g/dL Leukocyte count 13,500/mm³ Segmented neutrophils 75% Eosinophils 1% Lymphocytes 22% Monocytes 2% Platelet count 328,000/mm³ Erythrocyte sedimentation rate 82 mm/h Urine RBC 1–2/hpf WBC 0–1/hpf Nitrite negative Bacteria occasional Urine pregnancy test negative Which of the following is the most appropriate pharmacotherapy?"

Answer Choices:

A. Oral metronidazole B. Oral levofloxacin and azithromycin C. Oral trimethoprim-sulfamethoxazole D. Intramuscular ceftriaxone and oral doxycycline

Correct Answer: D. Intramuscular ceftriaxone and oral doxycycline

Question 10

Clinical Scenario:

A 29-year-old man is outside his home doing yard work when a bee stings him in the right arm. Within 10 minutes, he reports breathlessness and multiple, circular, pruritic rashes over his right arm. He drives to his family physician's office for evaluation. His past medical history is significant for hypertension and he takes lisinopril. Known allergies include latex, Hymenoptera, and aspirin. His blood pressure is 118/68 mm Hg; heart rate is 104/min and regular; respiratory rate is 22/min; temperature is 37.7°C (99.8°F). There is non-pitting edema but erythema with raised wheals are present in the region of the right arm. Auscultation of the lungs reveals mild wheezing at the lung bases. Which of the following is the best course of action in the management of this patient?

Answer Choices:

A. Diphenhydramine and go to the emergency department B. Methylprednisolone and go to the emergency department C. Go to the emergency department D. Epinephrine and go to the emergency department

Correct Answer: D. Epinephrine and go to the emergency department

Question 11

Clinical Scenario:

A medical research study is evaluating an investigational novel drug (medication 1) as compared with standard therapy (medication 2) in patients presenting to the emergency department with myocardial infarction (MI). The study enrolled a total of 3,000 subjects, 1,500 in each study arm. Follow-up was conducted at 45 days post-MI. The following are the results of the trial: Endpoints Medication 1 Medication 2 P-Value Primary: death from cardiac causes 134 210 0.03 Secondary: hyperkalemia 57 70 0.4 What is the relative risk of death from a cardiac cause? (Round to the nearest whole number.)

Answer Choices:

A. 42% B. 57% C. 64% D. 72%

Correct Answer: C. 64%

Question 12

Clinical Scenario:

A 42-year-old man comes to the physician because he is concerned that he is balding. Over the past few months, he has noticed patchy areas of hair loss on his head. He also mentions that he has felt depressed since the death of his wife last year and has unintentionally lost about 18 kg (40 lbs). He is constantly fatigued. He has little appetite because he feels food does not taste the same way anymore. He also has occasional episodes of watery diarrhea. He drinks 5–6 cans of beer daily. Vital signs are within normal limits. Examination shows dry, scaly skin on both feet. There is patchy alopecia of the scalp, axillae, chest, and mons pubis. Which of the following is most likely to directly improve this patient's alopecia?

Answer Choices:

A. Finasteride B. Griseofulvin C. Restriction of vitamin A-rich foods D. Zinc supplementation

Correct Answer: D. Zinc supplementation

Question 13

Clinical Scenario:

A 26-year-old man comes to the physician because of a 1-week history of left-sided chest pain. The pain is worse when he takes deep breaths. Over the past 6 weeks, he had been training daily for an upcoming hockey tournament. He does not smoke cigarettes or drink alcohol but has used cocaine once. His temperature is 37.1°C (98.7°F), pulse is 75/min, and blood pressure is 128/85 mm Hg. Physical examination shows tenderness to palpation of the left chest. An x-ray of the chest is shown. Which of the following is the most appropriate initial pharmacotherapy?

Answer Choices:

A. Alprazolam B. Alteplase C. Heparin D. Naproxen

Correct Answer: D. Naproxen

Question 14

Clinical Scenario:

A 12-year-old girl presents to a pediatrician because she fails to show signs of breast development. On physical examination, the pediatrician notes that her stature is shorter than expected for her age and sex. She has a webbed neck, a low posterior hairline, and a broad chest with widely spaced nipples. Non-pitting bilateral pedal edema is present. The pediatrician orders a karyotype analysis, the result of which is shown below. Which of the following findings is most likely to be present on auscultation of her chest?

Answer Choices:

A. A short systolic murmur along the left sternal border in the third and fourth intercostal spaces which radiates to the left infrascapular area B. A continuous machine-like murmur over the second left intercostal space which

radiates to the left clavicle C. A high-pitched holosystolic murmur over the apical area which radiates to the left axilla D. A loud and harsh holosystolic murmur which is heard best over the lower left sternal border

Correct Answer: A. A short systolic murmur along the left sternal border in the third and fourth intercostal spaces which radiates to the left infrascapular area

Question 15

Clinical Scenario:

A 28-year-old woman, gravida 2, para 1, at 14 weeks' gestation, comes to the physician with a 3-day history of abnormal vaginal discharge. She has not had fever, chills, or abdominal pain. One week ago, her 2-year-old daughter had a urinary tract infection that quickly resolved after antibiotic therapy. The patient reports that she is sexually active with one male partner and they do not use condoms. Vital signs are within normal limits. Pelvic examination shows an inflamed and friable cervix. There is mucopurulent, foul-smelling discharge from the cervical os. There is no uterine or cervical motion tenderness. Vaginal pH measurement shows a pH of 3.5. Which of the following is the most appropriate initial step in management?

Answer Choices:

A. Wet mount preparation B. Amine test C. Urine analysis and culture D. Nucleic acid amplification test

Correct Answer: D. Nucleic acid amplification test

Question 16

Clinical Scenario:

A 2-year-old boy is brought to the emergency department by his mother because of progressive fatigue, abdominal pain, and loss of appetite over the past 3 days. He was treated in the emergency department once in the past year for swelling of his hands and feet. He was adopted as a baby from Sudan and his family history is unknown. He does not take any medication. He is lethargic. His temperature is 37.5°C (99.5°F), pulse is 141/min, respirations are 25/min, and blood pressure is 68/40 mm Hg. Examination shows pale, dry mucous membranes and scleral icterus. Laboratory studies show: Hemoglobin 7.1 g/dL Mean corpuscular volume 93 fL Reticulocyte count 11% Serum Lactate dehydrogenase 194 IU/L Total bilirubin 6.4 mg/dL Direct bilirubin 0.5 mg/dL Haptoglobin 21 mg/dL (N = 41–165) Further evaluation of this patient is most likely to show which of the following findings?"

Answer Choices:

A. Anti-erythrocyte antibodies on Coombs test B. Splenomegaly on ultrasound C. Hypocellular bone marrow on biopsy D. Low ferritin level in serum

Correct Answer: B. Splenomegaly on ultrasound

Question 17

Clinical Scenario:

A 17-year-old man presents to his primary care physician with bilateral tremor of the hands. He is a senior in high school and during the year, his grades have plummeted to the point that he is failing. He says his memory is now poor, and he has trouble focusing on tasks. His behavior has changed in the past 6 months in that he has frequent episodes of depression, separated by episodes of bizarre behavior, including excessive alcohol drinking and shoplifting. His parents have started to suspect that he is using street drugs, which he denies. His handwriting has become very sloppy. His parents have noted slight slurring of his speech. Family history is irrelevant. Physical examination reveals upper extremity tremors, mild dystonia of the upper extremities, and mild incoordination involving his hands. The patient's eye is shown. Which of the following is the best initial management of this patient's condition?

Answer Choices:

A. Penicillamine B. Oral zinc C. Oral deferasirox D. Watchful waiting

Correct Answer: B. Oral zinc

Question 18

Clinical Scenario:

A 37-year-old man comes to the physician because of a 3-day history of fatigue and yellowish discoloration of his eyes and skin. Physical examination shows mild right upper quadrant abdominal tenderness. The course of different serum parameters over the following 4 months is shown. Which of the following is the most likely explanation for the course of this patient's laboratory findings?

Answer Choices:

A. Chronic hepatitis B infection with low infectivity B. Chronic hepatitis B infection with high infectivity C. Adverse reaction to hepatitis B vaccination D. Resolved acute hepatitis B infection

Correct Answer: D. Resolved acute hepatitis B infection

Question 19

Clinical Scenario:

A 28-year-old African American woman comes to the emergency department with intermittent and progressively worsening dizziness with near-fainting incidents for the last 3 weeks. She denies fever, weight loss, nausea, vomiting, or chest pain. Her medical history is significant for a chronic cough and intermittent skin rashes that spontaneously resolved after a few weeks. She does not smoke tobacco but drinks alcohol socially. The patient lives alone with no pets. Her temperature is 37°C (98.6°F), blood pressure is 122/80 mm Hg, pulse is 43/min, and respirations are 12/min. On physical examination, cervical lymphadenopathy is noted. No heart murmurs are heard. ECG shows sinus rhythm with 2:1 atrioventricular block and left bundle branch

block. Chest X-ray shows prominent hilar lymphadenopathy. Which of the following is the most likely diagnosis?

Answer Choices:

A. Carotid artery stenosis B. Non-Hodgkin's lymphoma C. Small cell lung cancer D. Sarcoidosis

Correct Answer: D. Sarcoidosis

Question 20

Clinical Scenario:

An 8-year-old girl is brought to the physician by her parents because of difficulty sleeping. One to two times per week for the past 2 months, she has woken up frightened in the middle of the night, yelling and crying. She has not seemed confused after waking up, and she is consolable and able to fall back asleep in her parents' bed. The following day, she seems more tired than usual at school. She recalls that she had a bad dream and looks for ways to delay bedtime in the evenings. She has met all her developmental milestones to date. Physical examination shows no abnormalities. Which of the following is the most likely diagnosis?

Answer Choices:

A. Sleep terror disorder B. Post-traumatic stress disorder C. Normal development D. Nightmare disorder

Correct Answer: D. Nightmare disorder

Question 21

Clinical Scenario:

A 24-year-old woman at 36 weeks pregnant presents to the emergency department with a headache and abdominal pain. The woman has no known past medical history and has inconsistently followed up with an obstetrician for prenatal care. Her temperature is 98.5°F (36.9°C), blood pressure is 163/101 mmHg, pulse is 90/min, respirations are 16/min, and oxygen saturation is 97% on room air. Prior to performing the physical exam, the patient experiences a seizure, which resolves after 60 seconds. Which of the following is the best management for this patient?

Answer Choices:

A. Diazepam, magnesium, and continuous monitoring B. Magnesium and continuous monitoring C. Magnesium and cesarean section D. Nifedipine and cesarean section

Correct Answer: C. Magnesium and cesarean section

Question 22

Clinical Scenario:

A 62-year-old man presents to the ED complaining of severe eye pain that started a few hours ago. The patient reports that he fell asleep while watching TV on the couch and woke up with right-sided eye pain and blurry vision. His wife drove him to the emergency room. His wife reports that since they arrived the patient has also been complaining of intense nausea. The patient denies fever, headache, or visual floaters. He has a history of hypertension, hyperlipidemia, type II diabetes mellitus, and osteoarthritis. He takes aspirin, lisinopril, metformin, atorvastatin, and over-the-counter ibuprofen. His temperature is 99°F (37.2°C), blood pressure is 135/82 mmHg, and pulse is 78/min. On physical examination, the right eye is firm with an injected conjunctiva and a mildly cloudy cornea. The pupil is dilated at 6 mm and is non-reactive to light. Ocular eye movements are intact. Vision is 20/200 in the right eye and 20/40 in the left eye. The left eye exam is unremarkable. Which of the following is the most appropriate initial treatment?

Answer Choices:

A. Intravenous acetazolamide B. Retinal photocoagulation C. Topical epinephrine D. Topical prednisolone

Correct Answer: A. Intravenous acetazolamide

Question 23

Clinical Scenario:

A 23-year-old woman presents to her primary care physician with 3 days of fatigue and back pain after she started a drug for malaria prophylaxis. She says that her urine has also been darker over the same time period. Her past medical history is significant for allergies as well as a broken elbow that was treated in a cast 10 years ago. She does not take any medications, does not smoke, and drinks socially. Peripheral blood smear reveals both red blood cells with dark intracellular inclusions as well as abnormally shaped red blood cells. The immune cells responsible for the shape of these red blood cells are located in which of the following places?

Answer Choices:

A. Bone marrow B. Blood vessels C. Lymph nodes D. Red pulp of the spleen

Correct Answer: D. Red pulp of the spleen

Question 24

Clinical Scenario:

A 54-year-old man is brought to the emergency department 1 hour after the sudden onset of shortness of breath, severe chest pain, and sweating. He has hypertension and type 2 diabetes mellitus. He has smoked one pack and a half of cigarettes daily for 20 years. An ECG shows ST-segment elevations in leads II, III, and avF. The next hospital with a cardiac catheterization unit is more than 2 hours away. Reperfusion pharmacotherapy is initiated. Which of the following is the primary mechanism of action of this medication?

Answer Choices:

A. Conversion of plasminogen to plasmin B. Prevention of thromboxane formation C. Inhibition of glutamic acid residue carboxylation D. Direct inhibition of thrombin activity

Correct Answer: A. Conversion of plasminogen to plasmin

Question 25

Clinical Scenario:

A 22-year-old woman comes to the physician for gradual worsening of her vision. Her father died at 40 years of age. She is 181 cm (5 ft 11 in) tall and weighs 69 kg (152 lb); BMI is 21 kg/m². A standard vision test shows severe myopia. Genetic analysis shows an FBN1 gene mutation on chromosome 15. This patient is at greatest risk of mortality due to which of the following causes?

Answer Choices:

A. Obstruction of the superior vena cava lumen B. Increased pressure in the pulmonary arteries C. Eccentric ventricular hypertrophy D. Intimal tear of the aortic root

Correct Answer: D. Intimal tear of the aortic root

Question 26

Clinical Scenario:

A 23-year-old man presents to the physician with nausea, vomiting, constipation, and abdominal pain for the past 24 hours. He has also developed difficulty in swallowing and blurring of vision. He also complains of significant dryness of his mouth and throat. When asked about his diet, he reports that he has been saving money by eating dented and old canned goods. On physical examination, his vital signs are stable. His neurologic examination reveals bilateral fixed dilated pupils, weakness of extraocular muscles, and weak gag reflex, while sensations and gait are normal. Laboratory evaluation of his stool isolates a toxin produced by gram-positive, anaerobic, spore-forming bacilli. Which of the following mechanisms best explains the action of the toxin?

Answer Choices:

A. Blockade of release of acetylcholine at neuromuscular junctions B. Inactivation of acetylcholinesterase at neuromuscular junctions C. Competitive antagonism of acetylcholine at postsynaptic receptors D. Prolonged depolarization of NM receptors

Correct Answer: A. Blockade of release of acetylcholine at neuromuscular junctions

Question 27

Clinical Scenario:

A 3-year-old girl swallowed a handful of pills after her grandmother dropped the bottle on the ground this afternoon. She presents to the ER in a very drowsy but agitated state. She is clutching her abdomen, as if in pain, her skin is dry and flushed, and she does not know her name or where she is. Her pupils are dilated. Her

grandmother reports that she has not urinated in several hours. The grandmother's medical history is significant for allergic rhinitis and osteoarthritis, both of which are treated with over the counter medications. What is the appropriate treatment for this child?

Answer Choices:

A. Atropine B. Naloxone C. Physostigmine D. Deferoxamine

Correct Answer: C. Physostigmine

Question 28

Clinical Scenario:

A 36-year-old woman is brought to the emergency department after being involved in a motor vehicle collision. She is alert, awake, and oriented. There is no family history of serious illness and her only medication is an oral contraceptive. Her temperature is 37.3°C (99°F), pulse is 100/min, respirations are 20/min, and blood pressure is 102/80 mm Hg. Physical examination shows ecchymoses over the trunk and abdomen. A FAST scan of the abdomen is negative. An x-ray of the chest shows no fractures. A contrast-enhanced CT scan of the chest and abdomen is performed that shows a 4-cm sharply defined liver mass with a hypoattenuated central scar. Which of the following is the most appropriate next step in management?

Answer Choices:

A. Reassurance and observation B. Biopsy of the mass C. Discontinue the oral contraceptive D. Percutaneous aspiration of the mass

Correct Answer: A. Reassurance and observation

Question 29

Clinical Scenario:

A 67-year-old man comes to the office due to pain in the lower part of his calves on his afternoon walk to get the mail. The pain is relieved by rest. It started slowly about 6 months ago and has become more painful over time. He has a history of hypertension, hyperlipidemia, diabetes mellitus, and smoking. Medications include hydrochlorothiazide, atorvastatin, metformin, and a multivitamin that he takes daily. The patient does not smoke and only drinks socially. Today, his blood pressure is 145/90 mm Hg, pulse is 75/min, respiratory rate is 17/min, and temperature is 37.6°C (99.6°F). On physical exam, he appears mildly obese and healthy. His heart has a regular rate and rhythm, and his lungs are clear to auscultation bilaterally. Examination of the legs shows atrophic changes and diminished pedal pulses. A measure of his ankle brachial index (ABI) is 0.89. Which of the following is the most appropriate initial treatment?

Answer Choices:

A. Enoxaparin B. Metoprolol C. A recommendation to perform pedal pumping exercises D. A referral to a supervised exercise program

Correct Answer: D. A referral to a supervised exercise program

Question 30

Clinical Scenario:

A 68-year-old woman is brought to the emergency department by ambulance after she was found down by her daughter. She lives alone in her apartment so it is unclear when she began to develop symptoms. Her medical history is significant for cardiac arrhythmias, diabetes, pericarditis, and a stroke 2 years ago. On presentation her temperature is 98.1°F (36.7°C), blood pressure is 88/51 mmHg, pulse is 137/min, and respirations are 18/min. On physical exam her skin is cold and clammy. If special tests were obtained, they would reveal dramatically decreased pulmonary capillary wedge pressure, increased systemic vascular resistance, and mildly decreased cardiac output. Which of the following treatments would most directly target the cause of this patient's low blood pressure?

Answer Choices:

A. Antibiotic administration B. Intravenous fluids C. Relieve obstruction D. Vasopressors

Correct Answer: B. Intravenous fluids

Question 31

Clinical Scenario:

A 35-year-old G2P0 presents to her physician to discuss the results of her 16-week obstetric screening tests. She has no complaints. Her previous pregnancy at 28 years of age was a spontaneous abortion in the first trimester. She has no history of gynecologic diseases. Her quadruple test shows the following findings: Alpha-fetoprotein Low Beta-hCG High Unconjugated estriol Low Inhibin A High Which of the following statements regarding the presented results is correct?

Answer Choices:

A. Such results are associated with a 100% lethal fetal condition. B. Maternal age is a significant risk factor for the condition of the patient, the increased risk of which is indicated by the results of the study. C. The obtained results can be normal for women aged 35 and older. D. The results show increased chances of aneuploidies associated with the sex chromosomes.

Correct Answer: B. Maternal age is a significant risk factor for the condition of the patient, the increased risk of which is indicated by the results of the study.

Question 32

Clinical Scenario:

A 47-year-old man with a history of alcoholism undergoes an upper endoscopy, which reveals a superficial mucosal tear in the distal esophagus. Laboratory results show a metabolic alkalosis. What is the most likely mechanism of the acid/base disturbance in this patient?

Answer Choices:

A. Anemia B. Vomiting C. Hypokalemia D. Hepatic cirrhosis

Correct Answer: B. Vomiting

Question 33**Clinical Scenario:**

A 5-week-old infant boy presents to the pediatrician with intermittent vomiting for the last 2 weeks. The mother reports that the vomiting is non-bilious and immediately follows feeding. After vomiting, the baby is hungry and wants to feed again. The frequency of vomiting has been increasing progressively over 2 weeks. The vital signs are within normal limits. The examination of the abdomen reveals the presence of a firm mass of approx. 2 cm in length, above and to the right of the umbilicus. The mass is movable, olive-shaped, and hard on palpation. Which of the following is the most likely surgical treatment for this infant's condition?

Answer Choices:

A. Surgical ligation of the fistula and primary end-to-end anastomosis of the esophagus B. Pyloromyotomy C. Duodenoduodenostomy D. Endorectal pull-through procedure

Correct Answer: B. Pyloromyotomy

Question 34**Clinical Scenario:**

Two days after delivery, a newborn develops a red, irritated eye with yellow discharge. She was born at 39 weeks' gestation to a 28-year-old woman, gravida 1, para 1. Pregnancy and delivery were uncomplicated. The mother had not seen her gynecologist since her first prenatal visit. The newborn's temperature is 37.2°C (99.0°F), pulse is 140/min, respirations are 42/min, and blood pressure is 73/53 mm Hg. Ophthalmic examination shows eyelid edema, conjunctival injection, and copious yellow mucopurulent discharge from the right eye. There is no corneal ulceration or evidence of keratitis. Funduscopy examination is normal. The diagnosis is confirmed and appropriate treatment is administered. Which of the following is most likely to have prevented this patient's condition?

Answer Choices:

A. IV ceftriaxone administered to the infant B. Topical erythromycin administered to the infant C. Oral doxycycline administered to the mother D. Oral amoxicillin administered to the mother

Correct Answer: B. Topical erythromycin administered to the infant

Question 35**Clinical Scenario:**

A 31-year-old woman comes to the physician because of headaches and nausea for 2 weeks. The headaches are worse on awakening and she describes them as 7 out of 10 in intensity. During this period, she has noticed brief episodes of visual loss in both eyes lasting several seconds, especially when she suddenly stands up or bends over. She is 165 cm (5 ft 5 in) tall and weighs 98 kg (216 lb); BMI is 36 kg/m². Vital signs are within normal limits. Examination shows a visual acuity of 20/20 in both eyes with mild peripheral vision loss. Fundoscopic examination shows bilateral optic disc swelling. An MRI of the brain shows no abnormalities. A lumbar puncture is performed; opening pressure is 310 mm H₂O. Cerebrospinal fluid analysis shows a leukocyte count of 4/mm³ (75% lymphocytes), a protein concentration of 35 mg/dL, and a glucose concentration of 45 mg/dL. Which of the following is the most appropriate next step in management?

Answer Choices:

A. Acetazolamide therapy B. Optic nerve sheath fenestration C. Ventricular shunting D. Furosemide therapy

Correct Answer: A. Acetazolamide therapy

Question 36

Clinical Scenario:

A previously healthy 20-year-old man comes to the physician because of a 6-month history of a painless mass in his left groin that has been gradually increasing in size. Physical examination shows a 3x3-cm oval, non-tender left inguinal mass and a fluctuant, painless left scrotal swelling that increase in size with coughing. Which of the following is the most likely cause of this patient's symptoms?

Answer Choices:

A. Failure of processus vaginalis to close B. Obstruction of left spermatic vein C. Widening of femoral ring D. Weakening of transversalis fascia "

Correct Answer: A. Failure of processus vaginalis to close

Question 37

Clinical Scenario:

A 62-year-old woman presents for a regular check-up. She complains of lightheadedness and palpitations which occur episodically. Past medical history is significant for a myocardial infarction 6 months ago and NYHA class II chronic heart failure. She also was diagnosed with grade I arterial hypertension 4 years ago. Current medications are aspirin 81 mg, atorvastatin 10 mg, enalapril 10 mg, and metoprolol 200 mg daily. Her vital signs are a blood pressure of 135/90 mm Hg, a heart rate of 125/min, a respiratory rate of 14/min, and a temperature of 36.5°C (97.7°F). Cardiopulmonary examination is significant for irregular heart rhythm and decreased S1 intensity. ECG is obtained and is shown in the picture (see image). Echocardiography shows a left ventricular ejection fraction of 39%. Which of the following drugs is the best choice for rate control in this patient?

Answer Choices:

A. Atenolol B. Diltiazem C. Propafenone D. Digoxin

Correct Answer: D. Digoxin

Question 38

Clinical Scenario:

A 55-year-old man presents with intense pain in his left knee that started after returning from a camping trip 2 days ago, during which he consumed copious amounts of alcohol and red meat. He says he has had similar episodes in the past that resolved spontaneously usually over a period of about 10 days. His past medical history is significant for essential hypertension managed with hydrochlorothiazide 20 mg/day. The patient is afebrile, and his vital signs are within normal limits. Physical examination shows edema, warmth, and erythema of the left knee, which is also severely tender to palpation; The range of motion at the left knee joint is limited. A joint arthrocentesis of the left knee is performed, and synovial fluid analysis reveals 20,000 neutrophils and the following image is seen under polarized light microscopy (see image). Which of the following is the best course of treatment for this patient's condition?

Answer Choices:

A. Nonsteroidal antiinflammatory drugs B. Uricosuric drug C. Intra-articular steroid injection D. Xanthine oxidase inhibitor

Correct Answer: A. Nonsteroidal antiinflammatory drugs

Question 39

Clinical Scenario:

An excisional biopsy is performed and the diagnosis of superficial spreading melanoma is confirmed. The lesion is 1.1 mm thick. Which of the following is the most appropriate next step in management?

Answer Choices:

A. Surgical excision with 0.5-1 cm safety margins only B. Surgical excision with 1-2 cm safety margins only
C. Surgical excision with 1-2 cm safety margins and sentinel lymph node study D. Surgical excision with 0.5-1 cm safety margins and sentinel lymph node study

Correct Answer: C. Surgical excision with 1-2 cm safety margins and sentinel lymph node study

Question 40

Clinical Scenario:

A 14-year-old boy is rushed to the emergency room after he became disoriented at home. His parents say that the boy was doing well until 2 days ago when he got sick and vomited several times. They thought he was recovering but today he appeared to be disoriented since the morning. His vitals are normal except shallow

rapid breathing at a rate of 33/min. His blood sugar level is 654 mg/dL and urine is positive for ketone bodies. He is diagnosed with diabetic ketoacidosis and is managed with fluids and insulin. He responds well to the therapy. His parents are told that their son has type 1 diabetes and insulin therapy options are being discussed. Which of the following types of insulin can be used in this patient for the rapid action required during mealtimes?

Answer Choices:

A. NPH insulin B. Insulin lispro C. Insulin glargine D. NPH and regular insulin

Correct Answer: B. Insulin lispro

Question 41

Clinical Scenario:

A 27-year-old woman comes to the physician because she has been hearing voices in her apartment during the past year. She also reports that she has been receiving warning messages in newspaper articles during this period. She thinks that “someone is trying to kill her”. She avoids meeting her family and friends because they do not believe her. She does not use illicit drugs. Physical examination shows no abnormalities. Mental status examination shows a normal affect. Which of the following is the most appropriate long-term treatment?

Answer Choices:

A. Quetiapine B. Fluphenazine C. Lithium carbonate D. Clozapine

Correct Answer: A. Quetiapine

Question 42

Clinical Scenario:

A 32-year-old man is brought to the emergency department 10 minutes after he sustained a stab wound to the left chest just below the clavicle. On arrival, he is hypotensive with rapid and shallow breathing and appears anxious and agitated. He is intubated and mechanically ventilated. Infusion of 0.9% saline is begun. Five minutes later, his pulse is 137/min and blood pressure is 84/47 mm Hg. Examination shows a 3-cm single stab wound to the left chest at the 4th intercostal space at the midclavicular line without active external bleeding. Cardiovascular examination shows muffled heart sounds and jugular venous distention. Breath sounds are normal bilaterally. Further evaluation of this patient is most likely to show which of the following findings?

Answer Choices:

A. A 15 mm Hg decrease in systolic blood pressure during inspiration B. Lateral shift of the trachea toward the right side C. Subcutaneous crepitus on palpation of the chest wall D. Inward collapse of part of the chest with inspiration

Correct Answer: A. A 15 mm Hg decrease in systolic blood pressure during inspiration

Question 43

Clinical Scenario:

A 37-year-old man presents to the physician. He has been overweight since childhood. He has not succeeded in losing weight despite following different diet and exercise programs over the past several years. He has had diabetes mellitus for 2 years and severe gastroesophageal reflux disease for 9 years. His medications include metformin, aspirin, and pantoprazole. His blood pressure is 142/94 mm Hg, pulse is 76/min, and respiratory rate is 14/min. His BMI is 36.5 kg/m². Laboratory studies show: Hemoglobin A1C 6.6% Serum Fasting glucose 132 mg/dL Which of the following is the most appropriate surgical management?

Answer Choices:

A. Biliopancreatic diversion and duodenal switch (BPD-DS) B. Laparoscopic adjustable gastric banding C. Laparoscopic Roux-en-Y gastric bypass D. No surgical management at this time

Correct Answer: C. Laparoscopic Roux-en-Y gastric bypass

Question 44

Clinical Scenario:

A 58-year-old woman presents to the physician's office with vaginal bleeding. The bleeding started as a spotting and has increased and has become persistent over the last month. The patient is G3P1 with a history of polycystic ovary syndrome and type 2 diabetes mellitus. She completed menopause 4 years ago. She took cyclic estrogen-progesterone replacement therapy for 1 year at the beginning of menopause. Her weight is 89 kg (196 lb), height 157 cm (5 ft 2 in). Her vital signs are as follows: blood pressure 135/70 mm Hg, heart rate 78/min, respiratory rate 12/min, and temperature 36.7°C (98.1°F). Physical examination is unremarkable. Transvaginal ultrasound reveals an endometrium of 6 mm thickness. Speculum examination shows a cervix without focal lesions with bloody discharge from the non-dilated external os. On pelvic examination, the uterus is slightly enlarged, movable, and non-tender. Adnexa is non-palpable. What is the next appropriate step in the management of this patient?

Answer Choices:

A. Hysteroscopy with dilation and curettage B. Endometrial biopsy C. Saline infusion sonography D. Hysteroscopy with targeted biopsy

Correct Answer: B. Endometrial biopsy

Question 45

Clinical Scenario:

A 23-year-old woman visits her general practitioner with left ear pain and fever. She complains of multiple episodes of respiratory infection including bronchitis, laryngitis, and sinusitis. She was diagnosed with systemic lupus erythematosus with nephritis 8 months ago and was placed on oral prednisone. Currently, she

takes prednisone daily. Her vital signs are as follows: blood pressure 130/85 mm Hg, heart rate 79/min, respiratory rate 16/min, and temperature 37.5°C (99.5°F). Her weight is 78 kg (172 lb) and height is 169 cm (5 ft 6 in). Physical examination reveals a swollen erythematous left eardrum, erythematous macular rash over sun-exposed skin, and slight calf edema. Inhibition of which of the following pathways causes diminished immune cell activation in this patient?

Answer Choices:

A. Wnt pathway B. NF- κ B pathways C. PI3K/AKT/mTOR pathway D. Notch pathway

Correct Answer: B. NF- κ B pathways

Question 46

Clinical Scenario:

A 37-year-old woman, G1P0, visits her gynecologist's office for a routine prenatal checkup. During her quadruple screening test, her alpha-fetoprotein levels were increased while the β -hCG and pregnancy-associated plasma protein were decreased. There is also evidence of increased nuchal translucency on the scanning of the male fetus. A confirmatory test indicates signs of a genetic syndrome. The woman is counseled that her child will most likely have a severe intellectual disability. Physical features of this condition include polydactyly, cleft palate, micrognathia and clenched fists. This genetic condition also affects the formation of the brain and can lead to stillbirth. Most babies do not survive beyond the first year of life. Which of the following is responsible for this type of genetic syndrome?

Answer Choices:

A. In utero infections B. Error in metabolism C. Nondisjunction of chromosomes D. Autosomal dominant genes

Correct Answer: C. Nondisjunction of chromosomes

Question 47

Clinical Scenario:

A 57-year-old woman with a history of diabetes and hypertension accidentally overdoses on antiarrhythmic medication. Upon arrival in the ER, she is administered a drug to counteract the effects of the overdose. Which of the following matches an antiarrhythmic with its correct treatment in overdose?

Answer Choices:

A. Quinidine and insulin B. Encainide and epinephrine C. Esmolol and glucagon D. Sotalol and norepinephrine

Correct Answer: C. Esmolol and glucagon

Question 48

Clinical Scenario:

A 59-year-old male presents to his primary care physician complaining of muscle weakness. Approximately 6 months ago, he started to develop gradually worsening right arm weakness that progressed to difficulty walking about three months ago. His past medical history is notable for a transient ischemic attack, hypertension, hyperlipidemia, and diabetes mellitus. He takes aspirin, lisinopril, atorvastatin, metformin, and glyburide. He does not smoke and he drinks alcohol occasionally. Physical examination reveals 4/5 strength in right shoulder abduction and right arm flexion. A tremor is noted in the right hand. Strength is 5/5 throughout the left upper extremity. Patellar reflexes are 3+ bilaterally. Sensation to touch and vibration is intact in the bilateral upper and lower extremities. Tongue fasciculations are noted. Which of the following is the most appropriate treatment in this patient?

Answer Choices:

A. Natalizumab B. Selegiline C. Bromocriptine D. Riluzole

Correct Answer: D. Riluzole

Question 49

Clinical Scenario:

A 30-year-old woman presents to her family doctor requesting sleeping pills. She is a graduate student and confesses that she is a “worry-a-holic,” which has been getting worse for the last 6 months as the due date for her final paper is approaching. During this time, she feels more on edge, irritable, and is having difficulty sleeping. She has already tried employing good sleep hygiene practices, including a switch to non-caffeinated coffee. Her past medical history is significant for depression in the past that was managed medically. No current medications. The patient’s family history is significant for her mother who has a panic disorder. Her vital signs are within normal limits. Physical examination reveals a mildly anxious patient but is otherwise normal. Which of the following is the most effective treatment for this patient’s condition?

Answer Choices:

A. Buspirone B. Bupropion C. Desensitization therapy D. Relaxation training

Correct Answer: A. Buspirone

Question 50

Clinical Scenario:

A 27-year-old man presents with a 2-week history of fever, malaise, and occasional diarrhea. On physical examination, the physician notes enlarged inguinal lymph nodes. An HIV screening test is positive. Laboratory studies show a CD4+ count of 650/mm³. This patient is most likely currently in which of the following stages of HIV infection?

Answer Choices:

A. Latent HIV infection B. AIDS C. Acute HIV infection D. Asymptomatic HIV infection

Correct Answer: C. Acute HIV infection

Question 51**Clinical Scenario:**

A 59-year-old male presents to his primary care physician complaining of a tremor. He developed a tremor in his left hand approximately three months ago. It appears to be worse at rest and diminishes if he points to something or uses the hand to hold an object. His past medical history is notable for emphysema and myasthenia gravis. He has a 40 pack-year smoking history. Physical examination reveals slowed movements. The patient takes several seconds to rise from his chair for a gait analysis which reveals a shuffling gait. The physician decides to start the patient on a medication that prevents the degradation of a neurotransmitter. This medication is also indicated for use in which of the following conditions?

Answer Choices:

A. Major depressive disorder B. Influenza C. Seasonal allergies D. Restless leg syndrome

Correct Answer: A. Major depressive disorder

Question 52**Clinical Scenario:**

A 67-year-old man presents to his primary care physician with constant and gnawing lower abdominal pain for 2 days. The pain has been steadily worsening in intensity. He says the pain occasionally radiates to his lower back and groin bilaterally. While he cannot identify any aggravating factors, he feels that the pain improves with his knees flexed. His medical history is notable for hypertension which is well controlled with medications. He has smoked 40–50 cigarettes daily for 35 years. On examination, there is a palpable pulsatile mass just left of midline below the umbilicus. He is immediately referred for definitive management but during transfer, he becomes hypotensive and unresponsive. Which of the following is the most likely diagnosis?

Answer Choices:

A. Gastrointestinal hemorrhage B. Ruptured abdominal aortic aneurysm C. Appendicitis D. Irritable bowel syndrome

Correct Answer: B. Ruptured abdominal aortic aneurysm

Question 53**Clinical Scenario:**

A 23-year-old woman comes to the physician because of a 3-month history of pain during intercourse and vaginal dryness. The patient has also had intermittent hot flashes and fatigue during this time. Over the past year, her periods have become irregular. Her last menstrual period was over six months ago. She is sexually active with one partner and does not use protection or contraception. She has a history of acute lymphoblastic leukemia during childhood, which has remained in remission. Pelvic examination shows an atrophic cervix and vagina. A urinary pregnancy test is negative. A progestin challenge test is performed and shows no withdrawal bleeding. Further evaluation of this patient is most likely to show which of the following findings?

Answer Choices:

A. Decreased GnRH levels B. Decreased LH levels C. Increased FSH to LH ratio D. Increased TSH levels

Correct Answer: C. Increased FSH to LH ratio

Question 54

Clinical Scenario:

A 3-year-old boy is brought to the emergency department after losing consciousness. His parents report that he collapsed and then had repetitive, twitching movements of the right side of his body that lasted approximately one minute. He recently started to walk with support. He speaks in bisyllables and has a vocabulary of almost 50 words. Examination shows a large purple-colored patch over the left cheek. One week later, he dies. Which of the following is the most likely finding on autopsy of the brain?

Answer Choices:

A. Intraparenchymal cyst B. Brainstem glioma C. Leptomeningeal vascular malformation D. Subependymal giant cell astrocytoma

Correct Answer: C. Leptomeningeal vascular malformation

Question 55

Clinical Scenario:

A 32-year-old woman, gravida 2, para 1, at 20 weeks' gestation comes to the physician for a prenatal visit. She feels well. Her first pregnancy was uncomplicated and the child was delivered vaginally. Medications include folic acid and an iron supplement. Her temperature is 37°C (98.6°F), pulse is 98/min, respirations are 18/min, and blood pressure is 108/76 mm Hg. Abdominal examination shows a uterus that is consistent with a 20-week gestation. The second-trimester scan shows no abnormalities. The patient intends to travel next month to Mozambique to visit her grandmother. Which of the following drugs is most suitable for pre-exposure prophylaxis against malaria?

Answer Choices:

A. Doxycycline B. Mefloquine C. Primaquine D. Proguanil

Correct Answer: B. Mefloquine

Question 56

Clinical Scenario:

A neurophysiologist describes the mechanism of a specific type of synaptic transmission to his students. While illustrating this, he points out that when the action potential reaches the presynaptic terminal of a chemical synapse, the voltage-gated Ca^{2+} channels open. Ca^{2+} ions trigger the release of neurotransmitters from vesicles in the presynaptic terminal. In this type of synaptic transmission, increased cytosolic Ca^{2+} levels cause the release of a neurotransmitter from small vesicles with dense cores. Which of the following neurotransmitters is most likely to be the one that is released into the synaptic cleft in this type of synapse?

Answer Choices:

A. Follicle stimulating hormone B. Epinephrine C. GABA (γ -amino butyric acid) D. Glutamate

Correct Answer: B. Epinephrine

Question 57

Clinical Scenario:

A 75-year-old man presents to the emergency department because of pain in his left thigh and left calf for the past 3 months. The pain occurs at rest, increases with walking, and is mildly improved by hanging the foot off the bed. He has had hypertension for 25 years and type 2 diabetes mellitus for 30 years. He has smoked 30–40 cigarettes per day for the past 45 years. On examination, femoral, popliteal, and dorsalis pedis pulses are faint on both sides. The patient's foot is shown in the image. Resting ankle-brachial index (ABI) is found to be 0.30. Antiplatelet therapy and aggressive risk factors modifications are initiated. Which of the following is the best next step for this patient?

Answer Choices:

A. Systemic anticoagulation with heparin B. Urgent assessment for revascularization C. Exercise therapy D. Amputation

Correct Answer: B. Urgent assessment for revascularization

Question 58

Clinical Scenario:

A 34-year-old man comes to the physician for a routine health maintenance examination. He was diagnosed with HIV 8 years ago. He is currently receiving triple antiretroviral therapy. He is sexually active and uses condoms consistently. He is planning a trip to Thailand with his partner to celebrate his 35th birthday in 6 weeks. His last tetanus and diphtheria booster was given 4 years ago. He received three vaccinations against hepatitis B 5 years ago. He had chickenpox as a child. Other immunization records are unknown. Vital signs are within normal limits. Cardiopulmonary examination shows no abnormalities. Leukocyte count shows

8,700/mm³, and CD4⁺ T-lymphocyte count is 480 cells/mm³ (Normal $\geq$ 500); anti-HBs is 150 mIU/mL. Which of the following recommendations is most appropriate at this time?

Answer Choices:

A. Bacillus Calmette Guerin vaccine B. Measles, mumps, rubella vaccine C. Yellow fever vaccine D. No vaccination

Correct Answer: B. Measles, mumps, rubella vaccine

Question 59

Clinical Scenario:

A 69-year-old man is brought to the emergency department because of severe epigastric pain and vomiting that started 30 minutes ago while gardening. His pulse is 55/min, respirations are 30/min, and blood pressure is 90/50 mm Hg. Physical examination shows diaphoresis and jugular venous distention. Crackles are heard in both lower lung fields. An ECG shows P waves independent of QRS complexes and ST segment elevation in leads II, III, and aVF. Coronary angiography is most likely to show narrowing of which of the following vessels?

Answer Choices:

A. Proximal right coronary artery B. Left circumflex artery C. Left anterior descending artery D. Posterior interventricular artery

Correct Answer: A. Proximal right coronary artery

Question 60

Clinical Scenario:

A 14-year-old boy presents to his pediatrician with weakness and frequent episodes of dizziness. He had chronic mucocutaneous candidiasis when he was 4 years old and was diagnosed with autoimmune hypoparathyroidism at age 8. On physical examination, his blood pressure is 118/70 mm Hg in the supine position and 96/64 mm Hg in the upright position. Hyperpigmentation is present over many areas of his body, most prominently over the extensor surfaces, elbows, and knuckles. His laboratory evaluation suggests the presence of antibodies to 21-hydroxylase and a mutation in the AIRE (autoimmune regulator) gene. The pediatrician explains to his parents that his condition is due to the failure of immunological tolerance. Which of the following mechanisms is most likely to have failed in the child?

Answer Choices:

A. Positive selection B. Negative selection C. Inhibition of the inactivation of harmful lymphocytes by regulatory T cells D. Deletion of mature lymphocytes

Correct Answer: B. Negative selection

Question 61

Clinical Scenario:

A 43-year-old man with a history of chronic alcoholism presents with a chronic cough and dyspnea. He says he traveled to Asia about 4 months ago and his symptoms started shortly after he returned. His temperature is 40.2°C (104.4°F) and pulse is 92/min. Physical examination reveals poor personal hygiene and a cough productive of foul blood-streaked sputum. Auscultation reveals decreased breath sounds on the right. A chest radiograph reveals an ill-defined circular lesion in the right middle lobe. Which of the following is true regarding this patient's most likely diagnosis?

Answer Choices:

A. Stains of gastric washing and urine have a high diagnostic yield on microscopy. B. Inoculation of a sputum sample into selective agar media needs to be incubated at 35–37°C (95.0–98.6°F) for up to 8 weeks. C. Ziehl-Neelsen staining is more sensitive than fluorescence microscopy with auramine-rhodamine stain. D. A positive tuberculin test would be diagnostic of active infection.

Correct Answer: B. Inoculation of a sputum sample into selective agar media needs to be incubated at 35–37°C (95.0–98.6°F) for up to 8 weeks.

Question 62

Clinical Scenario:

A 22-year-old female with no past medical history presents to her primary care physician with a 3-day history of knee pain. She denies any recent injury or trauma. On physical examination her knee is warm, erythematous, and has diminished range of movement. The patient reports to having multiple sexual partners over the last year and does not use protection regularly. Her blood pressure is 124/85 mmHg, heart rate is 76/min, and temperature is 38.3°C (101.0°F). A joint aspiration is performed and a growth of gram-negative diplococci is noted on bacterial culture. What is the treatment of choice for this patient's condition?

Answer Choices:

A. Nafcillin monotherapy and joint aspiration B. Oxacillin and ceftriaxone C. Vancomycin monotherapy D. Ceftriaxone monotherapy and joint aspiration

Correct Answer: D. Ceftriaxone monotherapy and joint aspiration

Question 63

Clinical Scenario:

A 22-year-old woman comes to the emergency department because of chest and epigastric pain that started just after vomiting 30 minutes ago. She does not take any medications and does not drink alcohol or smoke cigarettes. While in the emergency department, the patient experiences two episodes of forceful, bloody emesis. Her temperature is 99.1°F (37.3°C), pulse is 110/minute, and blood pressure is 105/60 mm Hg. Physical

examination shows dental enamel erosion and calluses on the dorsal aspect of her right hand. There is tenderness to palpation in the epigastrium. An x-ray of the chest is normal. Further evaluation of this patient is most likely to show which of the following findings?

Answer Choices:

A. Dilated veins in the esophageal submucosa B. Rupture of the distal esophagus C. Mucosal lacerations at the gastroesophageal junction D. Friable mass in the distal esophagus

Correct Answer: C. Mucosal lacerations at the gastroesophageal junction

Question 64

Clinical Scenario:

A 23-year-old man presents with a blunt force injury to the head from a baseball bat. He is currently unconscious, although his friends say he was awake and speaking with them en route to the hospital. He has no significant past medical history and takes no current medications. The vital signs include: temperature 37.0°C (98.6°F), blood pressure 165/85 mm Hg, pulse 50/min, and respiratory rate 19/min. On physical examination, there is a blunt force injury to the left temporoparietal region approximately 10.1–12.7 cm (4–5 in) in diameter. There is anisocoria of the left pupil, which is unresponsive to light. The patient is intubated and fluid resuscitation is initiated. A noncontrast computed tomography (CT) scan of the head is acquired and shown in the exhibit (see image). Which of the following is the most appropriate medical treatment for this patient?

Answer Choices:

A. Mannitol B. Maintain a PaCO₂ of 24 mm Hg C. Placement of a ventriculoperitoneal (VP) shunt D. Acetazolamide

Correct Answer: A. Mannitol

Question 65

Clinical Scenario:

A 64-year-old man is brought to the emergency department by his wife with a 2-hour history of diarrhea and vomiting. He says that he felt fine in the morning, but noticed that he was salivating, sweating, and feeling nauseated on the way home from his work as a landscaper. The diarrhea and vomiting then started about 10 minutes after he got home. His past medical history is significant for depression and drug abuse. His wife says that he has also been more confused lately and is afraid he may have ingested something unusual. Physical exam reveals miosis, rhinorrhea, wheezing, and tongue fasciculations. Which of the following treatments would most likely be effective for this patient?

Answer Choices:

A. Ammonium chloride B. Atropine C. Naloxone D. Sodium bicarbonate

Correct Answer: B. Atropine

Question 66

Clinical Scenario:

A 68-year-old woman is brought to the emergency department by her husband because of acute confusion and sudden weakness of her left leg that lasted for about 30 minutes. One hour prior to admission, she was unable to understand words and had slurred speech for about 15 minutes. She has type 2 diabetes mellitus and hypertension. She has smoked 1 pack of cigarettes daily for 30 years. Current medications include metformin and hydrochlorothiazide. Her pulse is 110/min and irregular; blood pressure is 135/84 mmHg. Examination shows cold extremities. There is a mild bruit heard above the left carotid artery. Cardiac examination shows a grade 2/6 late systolic ejection murmur that begins with a midsystolic click. Neurological and mental status examinations show no abnormalities. An ECG shows irregularly spaced QRS complexes with no discernible P waves. Doppler ultrasonography shows mild left carotid artery stenosis. A CT scan and diffusion-weighted MRI of the brain show no abnormalities. Which of the following treatments is most likely to prevent future episodes of neurologic dysfunction in this patient?

Answer Choices:

A. Enalapril B. Warfarin C. Alteplase D. Aortic valve replacement

Correct Answer: B. Warfarin

Question 67

Clinical Scenario:

A 65-year-old woman comes to the physician because of a 2-month history of intermittent bleeding from her vagina. She has no history of serious illness and takes no medications. Pelvic ultrasound shows a thickened endometrial stripe and a left adnexal mass. Endometrial biopsy shows a well-differentiated adenocarcinoma. Laboratory studies show increased levels of inhibin B. Which of the following is the most likely diagnosis?

Answer Choices:

A. Yolk sac tumor B. Granulosa cell tumor C. Immature teratoma D. Serous cystadenocarcinoma

Correct Answer: B. Granulosa cell tumor

Question 68

Clinical Scenario:

A 34-year-old male visits the clinic with complaints of intermittent diarrhea over the past 6 months. He has lost 6.8 kg (15 lb) over that time period. His frequent bowel movements are affecting his social life and he would like definitive treatment. Past medical history is significant for chronic type 2 diabetes that is well controlled with insulin. No other family member has a similar condition. He does not smoke tobacco and drinks alcohol only on weekends. Today, his vitals are within normal limits. On physical exam, he appears gaunt and anxious. His heart has a regular rate and rhythm and his lungs are clear to auscultation bilaterally. Additionally, the

patient has a red-purple rash on his lower abdomen, groin, and the dorsum of both hands. The rash consists of pruritic annular lesions. He is referred to a dermatologist for core biopsy which is consistent with necrolytic migratory erythema. Further workup reveals a large hormone secreting mass in the tail of his pancreas. Which of the following is the action of the hormone that is in excess in this patient?

Answer Choices:

A. Activation of glycogen synthase B. Inhibition of acetone production C. Inhibition of gluconeogenesis D. Stimulation of lipolysis

Correct Answer: D. Stimulation of lipolysis

Question 69

Clinical Scenario:

A 56-year-old woman is brought to the emergency department after falling on her outstretched hand. Her wrist is clearly deformed by fracture and is painful to palpation. Her wrist and finger motion is limited due to pain. After treatment and discharge, her final total cost is \$2500. Her insurance plan has a \$300 copay for emergency medical visits after the annual deductible of \$2000 is met and before 20% coinsurance. Previously this year, she had 2 visits to the emergency department for asthma attacks, which cost her \$350 and \$450. She has had no other medical costs during this period. Given that she has no previous balance due, which of the following must she pay out of pocket for her visit to the emergency department?

Answer Choices:

A. \$200 B. \$800 C. \$1200 D. \$1700

Correct Answer: D. \$1700

Question 70

Clinical Scenario:

A 68-year-old Caucasian male complains of severe headache and pain while chewing. Upon examination, he is found to have a left visual field deficit. Laboratory results show elevated erythrocyte sedimentation rate. Which of the following drugs would be the best choice for treatment of this patient?

Answer Choices:

A. Propranolol B. Prednisone C. Pilocarpine D. Clopidogrel

Correct Answer: B. Prednisone

Question 71

Clinical Scenario:

A 70-year-old man presents with fever, headache, and vomiting. He says that symptoms onset acutely 2 days ago and have not improved. He also reports associated weakness and chills. Past medical history is significant for occasional heartburn. His temperature is 39.4°C (103.0°F), the pulse rate is 124/min, the blood pressure is 130/84 mm Hg, and the respiratory rate is 22/min. On physical examination, there is significant nuchal rigidity. No signs of raised intracranial pressure are present. A lumbar puncture is performed and cerebrospinal fluid (CSF) analysis shows lymphocyte-dominant pleocytosis with increased CSF protein levels. Bacteriological culture of the CSF reveals the growth of *Listeria monocytogenes*. Which of the following antibiotics is the best choice for the treatment of this patient?

Answer Choices:

A. Ampicillin B. Ceftriaxone C. Chloramphenicol D. Vancomycin

Correct Answer: A. Ampicillin

Question 72

Clinical Scenario:

An 84-year-old man presents to the emergency department for a loss of consciousness. The patient states that he was using the bathroom when he lost consciousness and fell, hitting his head on the counter. The patient has a past medical history of diabetes, hypertension, obesity, factor V leiden, constipation, myocardial infarction, and vascular claudication. His current medications include lisinopril, atorvastatin, valproic acid, propranolol, insulin, metformin, and sodium docusate. The patient denies use of illicit substances. His temperature is 99.5°F (37.5°C), blood pressure is 167/98 mmHg, pulse is 90/min, respirations are 15/min, and oxygen saturation is 98% on room air. Physical exam reveals an elderly man sitting comfortably in his stretcher. Cardiac exam reveals a systolic murmur heard at the right upper sternal border that radiates to the carotids. Pulmonary exam reveals mild bibasilar crackles. Neurological exam reveals 5/5 strength in his upper and lower extremities with normal sensation. The patient's gait is mildly unstable. The patient is unable to give a urine sample in the emergency department and states that he almost fainted again when he tried to. Which of the following is the most likely diagnosis?

Answer Choices:

A. Postural hypotension B. Seizure C. Cardiac arrhythmia D. Situational syncope

Correct Answer: D. Situational syncope

Question 73

Clinical Scenario:

Two hours after undergoing a left femoral artery embolectomy, an obese 63-year-old woman has severe pain, numbness, and tingling of the left leg. The surgery was without complication and peripheral pulses were weakly palpable postprocedure. She has type 2 diabetes mellitus, peripheral artery disease, hypertension, and hypercholesterolemia. Prior to admission, her medications included insulin, enalapril, carvedilol, aspirin, and rosuvastatin. She appears uncomfortable. Her temperature is 37.1°C (99.3°F), pulse is 98/min, and blood

pressure is 132/90 mm Hg. Examination shows a left groin surgical incision. The left lower extremity is swollen, stiff, and tender on palpation. Dorsiflexion of her left foot causes severe pain in her calf. Femoral pulses are palpated bilaterally. Pedal pulses are weaker on the left side as compared to the right side. Laboratory studies show: Hemoglobin 12.1 Leukocyte count 11,300/mm³ Platelet count 189,000/mm³ Serum Glucose 222 mg/dL Creatinine 1.1 mg/dL Urinalysis is within normal limits. Which of the following is the most likely cause of these findings?"

Answer Choices:

A. Deep vein thrombosis B. Reperfusion injury C. Rhabdomyolysis D. Cholesterol embolism

Correct Answer: B. Reperfusion injury

Question 74

Clinical Scenario:

A 57-year-old man with type 2 diabetes mellitus comes to the physician for a follow-up evaluation. He was recently diagnosed with hyperlipidemia, for which he takes several medications. His serum total cholesterol concentration is 295 mg/dL and serum high-density lipoprotein concentration is 19 mg/dL (N: > 40 mg/dL). The physician prescribes an additional drug that decreases hepatic production of triglycerides and reduces the release of VLDL and LDL through the inhibition of diacylglycerol acyltransferase 2. This patient should be advised to do which of the following?

Answer Choices:

A. Take aspirin shortly before taking the new drug to reduce pruritus B. Schedule a follow-up appointment in 2 weeks to check serum creatine kinase levels C. Avoid smoking because of the new drug's increased risk of thrombosis D. Check blood glucose levels after taking the new drug to detect hypoglycemia

Correct Answer: A. Take aspirin shortly before taking the new drug to reduce pruritus

Question 75

Clinical Scenario:

A 33-year-old woman comes to the physician 1 week after noticing a lump in her right breast. Fifteen years ago, she was diagnosed with osteosarcoma of her left distal femur. Her father died of an adrenocortical carcinoma at the age of 41 years. Examination shows a 2-cm, firm, immobile mass in the lower outer quadrant of the right breast. A core needle biopsy of the mass shows adenocarcinoma. Genetic analysis in this patient is most likely to show a defect in which of the following genes?

Answer Choices:

A. KRAS B. TP53 C. PTEN D. Rb

Correct Answer: B. TP53

Question 76

Clinical Scenario:

A 55-year-old African American man presents to the emergency department with central chest pressure. His symptoms started the day before. The pain was initially intermittent in nature but has become constant and radiates to his jaw and left shoulder. He also complains of some difficulty breathing. The patient was diagnosed with essential hypertension a year ago, but he is not taking any medications for it. The patient denies smoking, alcohol, or drug use. Family history is unremarkable. His blood pressure is 230/130 mm Hg in both arms, the temperature is 36.9°C (98.4°F), and the pulse is 90/min. ECG shows diffuse T wave inversion and ST depression in lateral leads. Laboratory testing is significant for elevated troponin. Which of the following is the first-line antihypertensive agent for this patient?

Answer Choices:

A. Esmolol and intravenous nitroglycerin B. Fenoldopam C. Diazepam D. Hydralazine

Correct Answer: A. Esmolol and intravenous nitroglycerin

Question 77

Clinical Scenario:

A 46-year-old woman presents with palpitations, tremors, and anxiety. She says these symptoms have been present ever since a recent change in her diabetic medication. The most recent time she felt these symptoms, her blood glucose level was 65 mg/dL, and she felt better after eating a cookie. Which of the following is the mechanism of action of the drug most likely to have caused this patient's symptoms?

Answer Choices:

A. Block reabsorption of glucose in proximal convoluted tubule (PCT) B. Inhibition of α -glucosidase C. Blocking of the ATP-sensitive K⁺ channels D. Decreased hepatic gluconeogenesis

Correct Answer: C. Blocking of the ATP-sensitive K⁺ channels

Question 78

Clinical Scenario:

A 25-year-old man is brought to the emergency department by paramedics with a seizure lasting over 30 minutes. The patient's neighbors found him outside his apartment with all four limbs flailing and not responding to his name. No significant past medical history. On physical examination, the patient continues to be unresponsive and slightly cyanotic with irregular breathing. His teeth are clenched tightly. Intravenous glucose and an anticonvulsant medication are administered. Which of the following is the mechanism of action of the drug that was most likely administered to stop this patient's seizure?

Answer Choices:

A. Prolongation of chloride channel opening B. Increase in frequency of chloride channel opening C. Blockage of voltage-gated calcium channels D. Inactivation of sodium channels

Correct Answer: B. Increase in frequency of chloride channel opening

Question 79

Clinical Scenario:

A 6-year-old boy is brought to the emergency department due to a severe infection. Laboratory work shows leukocytosis of $60 \times 10^9/L$ with marked left shift, but no blast cells. The patient is febrile and dehydrated. The physician believes that this is a severe reaction to the infection and orders a leukocyte alkaline phosphatase (LAP) stain on a peripheral smear. The LAP score is elevated. Which of the following statements best describes an additional characteristic of the condition this child is suffering from?

Answer Choices:

A. Myeloblasts and promyelocytes are expected to be found. B. A blood count will contain band forms, metamyelocytes, and myelocytes. C. Chemotherapy is the treatment of choice. D. The patient may develop anemia secondary to infection.

Correct Answer: B. A blood count will contain band forms, metamyelocytes, and myelocytes.

Question 80

Clinical Scenario:

A 62-year-old man returns to his physician for a follow-up examination. During his last visit 1 month ago splenomegaly was detected. He has had night sweats for the past several months and has lost 5 kg (11 lb) unintentionally during this period. He has no history of a serious illness and takes no medications. The vital signs are within normal limits. The physical examination shows no abnormalities other than splenomegaly. The laboratory studies show the following: Hemoglobin 9 g/dL Mean corpuscular volume $95 \mu m^3$ Leukocyte count $12,000/mm^3$ Platelet count $260,000/mm^3$ Ultrasound shows a spleen size of 15 cm (5.9 in) and mild hepatomegaly. The peripheral blood smear shows teardrop-shaped and nucleated red blood cells (RBCs) and immature myeloid cells. The marrow is very difficult to aspirate but reveals hyperplasia of all 3 lineages. The tartrate-resistant acid phosphatase (TRAP) test is negative. Clonal marrow plasma cells are not seen. JAK-2 is positive. The cytogenetic analysis is negative for translocation between chromosomes 9 and 22. Which of the following is the most appropriate curative management in this patient?

Answer Choices:

A. Adriamycin, bleomycin, vinblastine, and dacarbazine (ABVD) B. Allogeneic bone marrow transplantation C. Imatinib mesylate D. Splenectomy

Correct Answer: B. Allogeneic bone marrow transplantation

Question 81

Clinical Scenario:

A 35-year-old woman is started on a new experimental intravenous drug X. In order to make sure that she is able to take this drug safely, the physician in charge of her care calculates the appropriate doses to give to this patient. Data on the properties of drug X from a subject with a similar body composition to the patient is provided below:

Weight: 100 kg Dose provided: 1500 mg Serum concentration 15 mg/dL Bioavailability: 1

If the patient has a weight of 60 kg and the target serum concentration is 10 mg/dL, which of the following best represents the loading dose of drug X that should be given to this patient?

Answer Choices:

A. 150 mg B. 300mg C. 450 mg D. 600 mg

Correct Answer: D. 600 mg

Question 82

Clinical Scenario:

A 69-year-old man with history of coronary artery disease necessitating angioplasty and stent placement presents to the ED due to fever, chills, and productive cough for one day. He is started on levofloxacin and admitted because of his comorbidity and observed tachypnea of 35 breaths per minute. He is continued on his home medications including aspirin, clopidogrel, metoprolol, and lisinopril. He cannot ambulate as frequently as he would like due to his immediate dependence on oxygen. What intervention should be provided for deep venous thrombosis prophylaxis in this patient while hospitalized?

Answer Choices:

A. Clopidogrel is sufficient; hold aspirin B. Aspirin and clopidogrel are sufficient C. Warfarin D. Low molecular weight heparin

Correct Answer: D. Low molecular weight heparin

Question 83

Clinical Scenario:

An 18-year-old woman presents to the medical clinic 6 days after her boyfriend's condom broke during sexual intercourse. The patient states "I do not wish to get pregnant at this point in my life." She has no other medical conditions and takes no prescription medications. Her family history is negative. She is a social drinker, drinking approx. 3–4 days every month. She is currently in a monogamous relationship with her boyfriend and she believes her boyfriend is monogamous as well. The heart rate is 104/min, and the blood pressure is 124/80 mm Hg. On physical examination, she appears tiresome and nervous. The heart auscultation is absent of murmur, and the lungs are clear to auscultation bilaterally. Her ovaries and uterus are palpable. Speculum exam

shows no signs of trauma and a closed cervical os. Based on her history and physical examination, which of the following management strategies would you recommend?

Answer Choices:

A. Ulipristal acetate B. Copper-IUD C. Levonorgestrel D. Ethinyl estradiol

Correct Answer: B. Copper-IUD

Question 84

Clinical Scenario:

A 52-year-old man with chronic alcoholism presents to an ambulatory medical clinic, where the hepatologist elects to perform comprehensive hepatitis B screening, in addition to several other screening and preventative measures. Given the following choices, which serologic marker, if positive, would indicate the patient has immunity to the hepatitis B virus?

Answer Choices:

A. HBsAg B. HBsAb C. HBcAb D. HBeAg

Correct Answer: B. HBsAb

Question 85

Clinical Scenario:

A 13-year-old boy with recently diagnosed schizophrenia presents with feelings of anxiety. The patient says that he has been having feelings of dread, especially since a friend of his has been getting bullied at school. He feels troubled by these feelings almost every day and makes it difficult for him to get ready to go to school. He also has been hallucinating worse lately. Past medical history is significant for schizophrenia diagnosed 1 year ago. Current medications are fluphenazine. The patient is afebrile and vital signs are within normal limits. Physical examination is unremarkable. Which of the following medications would most likely be a better course of treatment for this patient?

Answer Choices:

A. Chlorpromazine B. Fluoxetine C. Ziprasidone D. Alprazolam

Correct Answer: C. Ziprasidone

Question 86

Clinical Scenario:

The police are called to investigate a domestic disturbance. The neighbors report hearing a man shouting "I'm gonna kill you" for the past 30 minutes followed by occasional screaming. The house was only recently

occupied by its new owner, a middle-aged lawyer. The police were greeted at the door by a man holding a broomstick. When asked what the disturbance was about, he admitted to being extremely afraid of spiders and had come across one as he was unpacking. What would be the single best course of treatment for this patient?

Answer Choices:

A. Cognitive behavioral therapy B. Anxiolytics C. Antidepressants D. Beta-blockers

Correct Answer: A. Cognitive behavioral therapy

Question 87

Clinical Scenario:

A 45-year-old man is rushed to the emergency department by his wife after complaining of sudden onset, an excruciating headache that started about an hour ago. On further questioning, the patient's wife gives a prior history of flank pain, hematuria, and hypertension in the patient, and she recalls that similar symptoms were present in his uncle. On examination, his GCS is 12/15, and when his hip joint and knee are flexed, he resists the subsequent extension of the knee. When the neck is flexed there is severe neck stiffness and it causes a patient's hips and knees to flex. During the examination, he lapses into unconsciousness. Which of the following mechanisms best explains what led to this patient's presentation?

Answer Choices:

A. Increased wall tension within an aneurysm B. Intracerebral hemorrhage due to vascular malformations C. Meningeal irritation from a space occupying lesion D. Uremic encephalopathy from chronic renal disease

Correct Answer: A. Increased wall tension within an aneurysm

Question 88

Clinical Scenario:

A 27-year-old man presents to the emergency room with persistent fever, nausea, and vomiting for the past 3 days. While waiting to be seen, he quickly becomes disoriented and agitated. Upon examination, he has visible signs of difficulty breathing with copious oral secretions and generalized muscle twitching. The patient's temperature is 104°F (40°C), blood pressure is 90/64 mmHg, pulse is 88/min, and respirations are 18/min with an oxygen saturation of 90% on room air. When the nurse tries to place a nasal cannula, the patient becomes fearful and combative. The patient is sedated and placed on mechanical ventilation. Which of the following is a risk factor for the patient's most likely diagnosis?

Answer Choices:

A. Contaminated beef B. Epiglottic cyst C. Mosquito bite D. Spelunking

Correct Answer: D. Spelunking

Question 89

Clinical Scenario:

A 44-year-old woman presents to her physician's office for weakness. She reports having some difficulty placing books on a high shelf and getting up from a seated position. She denies muscle pain or any new rashes. She has noticed a tremor that is worse with action and has been having trouble falling asleep and staying asleep. She has lost approximately 10 pounds unintentionally over the course of 2 months. Medical history is significant for type I diabetes mellitus managed with an insulin pump. Family history is notable for systemic lupus erythematosus in her mother and panic disorder in the father. Her temperature is 98.6°F (37 °C), blood pressure is 140/85 mmHg, pulse is 102/min, and respirations are 17/min. On physical exam, she is mildly diaphoretic and restless, she has notable lid retraction, and her hair is thin. She has 4/5 strength in the proximal upper and lower extremities. Biceps and patellar tendon reflexes are 3+. Which of the following laboratory findings are most likely present in this patient?

Answer Choices:

A. Anti-Mi-2 antibody positivity B. Decreased thyroid-stimulating hormone C. Increased erythrocyte sedimentation rate D. Normal laboratory results

Correct Answer: B. Decreased thyroid-stimulating hormone

Question 90

Clinical Scenario:

A 50-year-old man is brought to the emergency department 30 minutes after the sudden onset of severe pain in the middle of his chest. He describes the pain as tearing in quality; it radiates to his jaw. He is sweating profusely. He has a 5-year history of hypertension and was diagnosed with chronic bronchitis 3 years ago. He has smoked one pack of cigarettes daily for the past 33 years. Current medications include enalapril and formoterol. The patient appears agitated. His pulse is 104/min, and respirations are 26/min. Blood pressure is 154/98 mm Hg in his right arm and 186/108 mm Hg in his left arm. An x-ray of the chest shows moderate hyperinflation; the mediastinum has a width of 9 cm. An ECG shows no abnormalities. This patient is at increased risk of developing which of the following?

Answer Choices:

A. Tactile crepitus over the neck B. Muffled heart sounds C. Early diastolic knocking sound D. Diminished breath sounds over the right lung base

Correct Answer: B. Muffled heart sounds

Question 91

Clinical Scenario:

A 68-year-old woman, otherwise healthy, is admitted to the coronary care unit due to acute ischemic cardiomyopathy. No other significant past medical history. Her vital signs include: pulse 116/min, respiratory rate 21/min, temperature 37.4°C (99.3°F), and blood pressure 160/100 mm Hg. On physical examination, the patient is in distress. Cardiopulmonary exam is positive for bilateral pulmonary crackles at the lung bases, tachycardia, and jugular venous distension. Her laboratory findings are significant for a hemoglobin of 7.8 g/dL. She is initially treated with oxygen, antiplatelet therapy, nitroglycerin, and beta-blockers. In spite of these treatments, her angina does not subside. The patient is not a candidate for percutaneous coronary intervention, so she is being prepared for a coronary artery bypass graft. Which of the following would be the next, best step in management of this patient?

Answer Choices:

A. Transfuse packed red blood cells B. Transfuse whole blood C. Treat with erythropoietin D. Observation and supportive care

Correct Answer: A. Transfuse packed red blood cells

Question 92

Clinical Scenario:

An 11-year-old boy is brought to the clinic by his parents for poor academic performance. The patient's parents say that his teacher told them that he may have to repeat a grade because of his lack of progress, as he does not pay attention to the lessons, tends to fidget about in his seat, and often blurts out comments when it is someone else's turn to speak. Furthermore, his after-school karate coach says the patient no longer listens to instructions and has a hard time focusing on the activity at hand. The patient has no significant past medical history and is currently not on any medications. The patient has no known learning disabilities and has been meeting all developmental milestones. The parents are vehemently opposed to using any medication with a potential for addiction. Which of the following medications is the best course of treatment for this patient?

Answer Choices:

A. Diazepam B. Atomoxetine C. Methylphenidate D. Olanzapine

Correct Answer: B. Atomoxetine

Question 93

Clinical Scenario:

A 40-year-old man with alcohol use disorder is brought to the emergency department because of sudden-onset blurry vision, severe upper abdominal pain, and vomiting that started one day after he drank a bottle of paint thinner. Physical examination shows epigastric tenderness without rebound or guarding. Ophthalmologic examination shows a visual acuity of 20/200 bilaterally despite corrective lenses. Arterial blood gas analysis on room air shows: pH 7.21 Sodium 135 mEq/L Chloride 103 mEq/L Bicarbonate 13 mEq/L An antidote with which of the following mechanisms of action is the most appropriate therapy for this patient's condition?"

Answer Choices:

A. Activation of acetyl-CoA synthetase B. Inhibition of acetaldehyde dehydrogenase C. Inhibition of acetyl-CoA synthetase D. Inhibition of alcohol dehydrogenase

Correct Answer: D. Inhibition of alcohol dehydrogenase

Question 94**Clinical Scenario:**

A 2-month-old girl is admitted to the hospital because of a 1-day history of fever and difficulty breathing. She has also had nasal congestion for 2 days. She was born at 28 weeks' gestation and weighed 1105 g (2 lb 7 oz); she currently weighs 2118 g (4 lb 11 oz). Her neonatal course was complicated by respiratory distress syndrome. She required supplemental oxygen for 36 days following birth. She was diagnosed with bronchopulmonary dysplasia 3 weeks ago. The infant missed an appointment with the pediatrician 2 weeks ago. Her only medication is vitamin D drops. She appears lethargic. Her temperature is 38.6°C (101.4°F), pulse is 160/min, respirations are 55/min, and blood pressure is 80/45 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 87%. Physical examination shows moderate subcostal retractions. Wheezing is heard on auscultation of the chest. Her hemoglobin concentration is 10.5 g/dL, leukocyte count is 13,000/mm³, and platelet count is 345,000/mm³. Mechanic ventilatory support is initiated. After 4 days in the pediatric intensive care unit, the patient dies. Administration of which of the following is most likely to have prevented this patient's outcome?

Answer Choices:

A. Postnatal glucocorticoid B. Ceftriaxone C. Respiratory syncytial virus immune globulin D. Palivizumab

Correct Answer: D. Palivizumab

Question 95**Clinical Scenario:**

An infectious disease investigator is evaluating the diagnostic accuracy of a new interferon-gamma-based assay for diagnosing tuberculosis in patients who have previously received a Bacillus Calmette-Guérin (BCG) vaccine. Consenting participants with a history of BCG vaccination received an interferon-gamma assay and were subsequently evaluated for tuberculosis by sputum culture. Results of the study are summarized in the table below.

	Tuberculosis, confirmed by culture	No tuberculosis	Total
Positive interferon-gamma assay	90	6	96
Negative interferon-gamma assay	10	194	204
Total	100	200	300

Based on these results, what is the sensitivity of the interferon-gamma-based assay for the diagnosis of tuberculosis in this study?"

Answer Choices:

A. 194/200 B. 90/100 C. 90/96 D. 194/204

Correct Answer: B. 90/100

Question 96

Clinical Scenario:

A 42-year-old man presents to a free dermatology clinic, complaining of itchy skin over the past several days. He has no insurance and lives in a homeless shelter. The patient has no significant medical history. Physical evaluation reveals 2 mm erythematous papules and vesicles on his back and groin, with linear excoriation marks. Careful observation of his hands reveals serpiginous, grayish, threadlike elevations in the superficial epidermis, ranging from 3–9 mm in length in the webbing between several digits. What should be the suggested treatment in this case?

Answer Choices:

A. No medication should be administered, only proper hygiene. B. Antiviral medication C. Permethrin D. Broad-spectrum antibiotic

Correct Answer: C. Permethrin

Question 97

Clinical Scenario:

A 29-year-old woman is brought to the emergency department after an episode of syncope. For the past 10 days, she has had dyspnea and palpitations occurring with mild exertion. The patient returned from a hiking trip in Upstate New York 5 weeks ago. Except for an episode of flu with fever and chills a month ago, she has no history of serious illness. Her temperature is 37.3°C (99.1°F), pulse is 45/min, respirations are 21/min, and blood pressure is 148/72 mm Hg. A resting ECG is shown. Two-step serological testing confirms the diagnosis. Which of the following is the most appropriate treatment?

Answer Choices:

A. Intravenous ceftriaxone B. Oral doxycycline C. Atropine D. Permanent pacemaker implantation

Correct Answer: A. Intravenous ceftriaxone

Question 98

Clinical Scenario:

A 10-year-old boy from Sri Lanka suffers from an autosomal dominant condition, the hallmark of which is hyperimmunoglobulinemia E and eosinophilia. He suffers from recurrent infections and takes antibiotic chemoprophylaxis. A STAT3 mutation analysis has been performed to confirm the diagnosis of Job syndrome. Eosinophilia Eczema Hay fever Atopic dermatitis Recurrent skin and lung infections Bronchial asthma What combination of symptoms above is characteristic of this condition?

Answer Choices:

A. I, II, III B. I, II, V C. I, II, IV, V D. IV, V, VI

Correct Answer: B. I, II, V

Question 99

Clinical Scenario:

A 5-year-old boy is brought to the emergency department by his parents for difficulty breathing. He was playing outside in the snow and had progressive onset of wheezing and gasping. His history is notable for eczema and nut allergies. The patient has respirations of 22/min and is leaning forward with his hands on his legs as he is seated on the table. Physical examination is notable for inspiratory and expiratory wheezes on exam. A nebulized medication is started and begins to relieve his breathing difficulties. Which of the following is increased in this patient as a result of this medication?

Answer Choices:

A. Cyclic GMP B. Cyclic AMP C. Protein kinase C D. ATP

Correct Answer: B. Cyclic AMP

Question 100

Clinical Scenario:

An 84-year-old woman is brought to the physician by her son after he found her trying to hang herself from the ceiling because she felt that she was a burden to her family. Her family says that for the past 2 months she has had no energy to leave her room, has been sleeping most of the day, has lost 10 kg (22 lb), and cries every day. She was diagnosed with breast cancer that has metastasized to the liver 4 months ago. She moved in with her son and daughter-in-law shortly after the diagnosis. She initially underwent chemotherapy but discontinued the treatment when the metastases spread to the spine and brain. Her life expectancy is 1–2 weeks and she is currently receiving home-hospice care. Her only current medication is a fentanyl patch. She is 160 cm (5 ft 3 in) tall and weighs 46 kg (101.4 lb); BMI is 18 kg/m². Her vital signs are within normal limits. Examination shows slow speech, a flat affect, and depressed mood. Which of the following treatments is initially most likely to provide the greatest benefit for this patient?

Answer Choices:

A. Methylphenidate B. Electroconvulsive therapy C. Fluoxetine D. Bupropion

Correct Answer: A. Methylphenidate

Question 101

Clinical Scenario:

An 8-year-old girl is brought to the physician by her parents for a 10-month history of disturbing dreams and daytime sleepiness. She has difficulty falling asleep and says she sometimes sees ghosts just before falling

asleep at night. She has had a 7-kg (15-lb) weight gain during this period despite no changes in appetite. She is alert and oriented, and neurologic examination is unremarkable. During physical examination, she spontaneously collapses after the physician drops a heavy book, producing a loud noise. She remains conscious after the collapse. Polysomnography with electroencephalogram is most likely to show which of the following?

Answer Choices:

A. Periodic sharp waves B. Slow spike-wave pattern C. Rapid onset of beta waves D. Decreased delta wave sleep duration

Correct Answer: C. Rapid onset of beta waves

Question 102

Clinical Scenario:

A 49-year-old man comes to the emergency department because of recurrent abdominal pain for 1 week. The pain is worse after eating and he has vomited twice during this period. He was hospitalized twice for acute pancreatitis during the past year; the latest being 2 months ago. There is no family history of serious illness. His only medication is a vitamin supplement. He has a history of drinking five beers a day for several years but quit 1 month ago. His temperature is 37.1°C (98.8°F), pulse is 98/min and blood pressure 110/70 mm Hg. He appears uncomfortable. Examination shows epigastric tenderness to palpation; there is no guarding or rebound. A CT scan of the abdomen shows a 6-cm low attenuation oval collection with a well-defined wall contiguous with the body of the pancreas. Which of the following is the most appropriate next step in management?

Answer Choices:

A. Magnetic resonance cholangiopancreatography B. CT-guided percutaneous drainage C. Middle segment pancreatectomy D. Distal pancreatectomy

Correct Answer: B. CT-guided percutaneous drainage

Question 103

Clinical Scenario:

A 45-year-old male presents to his primary care physician for complaints of dizziness. The patient reports he experiences room-spinning dizziness lasting several hours at a time, approximately 2-3 times a month, starting 3 months ago. Upon questioning, the patient also reports right sided diminished hearing, tinnitus, and a sensation of ear fullness. Her temperature is 99 deg F (37.2 deg C), pulse 70/min, respirations 12, blood pressure 130 mmHg/85 mmHg, SpO2 99%. You decide to order an audiometric evaluation. What is the most likely finding of the audiogram?

Answer Choices:

A. Low frequency sensorineural hearing loss B. High frequency sensorineural hearing loss C. Low frequency conductive hearing loss D. Normal audiogram

Correct Answer: A. Low frequency sensorineural hearing loss

Question 104

Clinical Scenario:

An 11-year-old boy is brought to the physician for the evaluation of frequent falling. His mother reports that the patient has had increased difficulty walking over the last few months and has refused to eat solid foods for the past 2 weeks. He has met all developmental milestones. The patient has had multiple ear infections since birth. His temperature is 37°C (98.6°F), pulse is 90/min, and blood pressure is 120/80 mm Hg. Examination shows foot inversion with hammertoes bilaterally. His gait is wide-based with irregular and uneven steps. Laboratory studies show a serum glucose concentration of 300 mg/dL. Further evaluation of this patient is most likely to show which of the following findings?

Answer Choices:

A. Expansion of GAA trinucleotide repeats B. Absence of dystrophin protein C. Duplication of PMP22 gene D. Defect of ATM protein "

Correct Answer: A. Expansion of GAA trinucleotide repeats

Question 105

Clinical Scenario:

A 3-year-old boy is brought to the emergency department by his mother for abdominal pain. She states that he has refused to eat and keeps clutching his stomach saying "ow." She reports that he has not had any vomiting or diarrhea. She says that he has not had a bowel movement in 3 days. The family recently moved from Namibia and has not established care. He has no known medical conditions and takes no medications. The mother says there is a family history of a "blood illness." On physical examination, there is mild distension with tenderness in the bilateral lower quadrants without organomegaly. An ultrasound of the abdomen reveals 2 gallstones without gallbladder wall thickening or ductal dilation and a negative Murphy sign. An abdominal radiograph shows moderate stool burden in the large bowel and rectum. Labs are obtained, as below:

Hemoglobin: 9 g/dL Platelet count: 300,000/mm³ Mean corpuscular volume (MCV): 85 µm³ Reticulocyte count: 5% Lactate dehydrogenase (LDH): 532 U/L Leukocyte count: 11/mm³ Serum iron: 140 mcg/dL Transferrin saturation: 31% (normal range 20-50%) Total iron binding capacity (TIBC): 400 mcg/dL (normal range 240 to 450 mcg/dL)

A hemoglobin electrophoresis shows hemoglobin S, increased levels of hemoglobin F, and no hemoglobin A. The results are discussed with the patient's mother including recommendations for increasing fiber in the patient's diet and starting hydroxyurea. Which of the following should also be part of management for the patient's condition?

Answer Choices:

A. Iron supplementation B. Penicillin until age 5 C. Ursodeoxycholic acid D. Vaccination for parvovirus

Correct Answer: B. Penicillin until age 5

Question 106

Clinical Scenario:

A newborn infant is born at 40 weeks gestation to a G1P1 mother. The pregnancy was uncomplicated and was followed by the patient's primary care physician. The mother has no past medical history and is currently taking a multi-vitamin, folate, B12, and iron. The infant is moving its limbs spontaneously and is crying. His temperature is 98.7°F (37.1°C), blood pressure is 60/38 mmHg, pulse is 150/min, respirations are 33/min, and oxygen saturation is 99% on room air. Which of the following is the best next step in management?

Answer Choices:

A. Fluid resuscitation B. Intramuscular (IM) vitamin K and topical erythromycin C. No further management needed D. Silver nitrate eye drops and basic lab work

Correct Answer: B. Intramuscular (IM) vitamin K and topical erythromycin

Question 107

Clinical Scenario:

A 28-year-old man presents with a complaint of penile discharge. He says that he noticed a yellowish watery discharge from his penis since last week. He adds that he has painful urination only in the mornings, but he sometimes feels a lingering pain in his genital region throughout the day. He denies any fever, body aches, or joint pains. No significant past medical history or current medications. When asked about his social history, he mentions that he has regular intercourse with women he meets in bars, however, he doesn't always remember to use a condom. Physical examination is unremarkable. The penile discharge is collected and sent for analysis. Ceftriaxone IM is administered, after which the patient is sent home with a prescription for an oral medication. Which of the following oral drugs was most likely prescribed to this patient?

Answer Choices:

A. Ampicillin B. Doxycycline C. Gentamicin D. Streptomycin

Correct Answer: B. Doxycycline

Question 108

Clinical Scenario:

A 12-year-old boy is brought to the emergency department after a motor vehicle collision. He was being carpooled to school by an intoxicated driver and was involved in a high velocity head-on collision. The patient is otherwise healthy and has no past medical history. His temperature is 99.2°F (37.3°C), blood pressure is 80/45 mmHg, pulse is 172/min, respirations are 36/min, and oxygen saturation is 100% on room air. A FAST exam demonstrates free fluid in Morrison pouch. The patient's parents arrive and state that they are Jehovah's witnesses. They state they will not accept blood products for their son but will allow him to go to the operating room to stop the bleeding. Due to poor understanding and a language barrier, the parents are also refusing IV

fluids as they are concerned that this may violate their religion. The child is able to verbalize that he agrees with his parents and does not want any treatment. Which of the following is the best next treatment for this patient?

Answer Choices:

A. Blood products and emergency surgery B. IV fluids alone as surgery is too dangerous without blood product stabilization C. IV fluids and vasopressors followed by emergency surgery D. Observation and monitoring and obtain a translator

Correct Answer: A. Blood products and emergency surgery

Question 109

Clinical Scenario:

An investigator is studying muscle contraction in tissue obtained from the thigh muscle of an experimental animal. After injection of radiolabeled ATP, the tissue is stimulated with electrical impulses. Radioassay of these muscle cells is most likely to show greatest activity in which of the following structures?

Answer Choices:

A. H zone B. I band C. A band D. Z line

Correct Answer: C. A band

Question 110

Clinical Scenario:

A 52-year-old unconscious man is brought to the emergency department. He was found unresponsive on the sidewalk in the snow. He is recognized by the staff as a local homeless man and IV drug user. Rapid warming procedures are initiated. At physical examination, he is dirty and disheveled and unrousable with a blood pressure of 100/76 mm Hg and a temperature of 37.2°C (99°F). He is thin with apparent weight loss. Both arms have indications of recent IV injection stigmata. A head MRI reveals multiple hyperintense signals in the meninges with multiple tiny contrast-enhancing lesions in the cerebellum and cerebral cortex. A chest X-ray is within normal limits. Mild dilatation of the ventricles is also appreciated. Cerebrospinal analysis fluid (CSF) analysis reveals: CSF opening pressure 25 cm H₂O CSF total leukocyte count 580/mm³ Lymphocytes 90% Neutrophils 10% CSF protein 176 mg/dL CSF glucose 21 mg/dL A specimen stains are positive for acid-fast bacilli. CSF culture is pending. Appropriate antibacterial medication is initiated. Which of the following is true regarding the immediate future management of this patient?

Answer Choices:

A. Acyclovir should be started empirically as well B. Check liver enzymes regularly C. Verify response to antibiotic therapy D. Treatment should only be started after CSF culture results

Correct Answer: B. Check liver enzymes regularly

Question 111

Clinical Scenario:

An 82-year-old woman is brought to the physician by her nephew, who lives with her because she has a pessimistic attitude and has displayed overall distrust of her nephew for 1 year. She frequently argues with her nephew and embarrasses him in front of his friends. She had a Colles' fracture 2 months ago and has had hypertension for 18 years. Her medications include hydrochlorothiazide and nortriptyline. She has a quantity of each leftover since her previous visit 2 months ago and has not requested new prescriptions, which she would need if she were taking them as prescribed. She appears untidy. Her blood pressure is 155/98 mm Hg. She mumbles in response to questions, and her nephew insists on being at her side during the entire visit because she cannot express herself clearly. She has a sore on her ischial tuberosity and bruises around her ankles. Which of the following is the most appropriate action in patient care?

Answer Choices:

A. Discussing advance directives B. Emphasizing compliance with medication and follow-up in 1 month C. Referral for hospice care D. Reporting possible elder abuse by phone

Correct Answer: D. Reporting possible elder abuse by phone

Question 112

Clinical Scenario:

A 17-year-old girl is brought to the physician by her mother for evaluation of mild acne. Six months ago, the girl developed papules over her back and shoulders. Her mother reports that her daughter has only been wearing clothes that cover her complete back and shoulders recently and that she spends a lot of time checking her skin in the mirror. She spends three hours a day scratching and squeezing the comedones. After reading an article that suggested sugar was a possible cause of acne, she tried a low-carb diet, which resulted in a weight loss 5.2-kg (11.5-lb) but no change in her skin condition. The patient describes herself as "ugly." Over the past 6 months, she quit the swim team, stopped swim training, and stayed home from school on several occasions. She appears sad and distressed. She is 170 cm (5 ft 7 in) tall and weighs 62 kg (136.7 lb); BMI is 21.4 kg/m². Vital signs are within normal limits. Physical examination shows a few small papules but numerous, widespread scratch marks over the neck, back, and buttocks. On mental status examination, she is depressed and irritable. There is no evidence of suicidal ideation. After establishing a therapeutic alliance, which of the following is the most appropriate next step in management?

Answer Choices:

A. Dialectical behavioral therapy B. Suggest hospitalization C. Nutritional rehabilitation D. Cognitive-behavioral therapy

Correct Answer: D. Cognitive-behavioral therapy

Question 113

Clinical Scenario:

A 24-year-old woman comes to the physician because of progressively worsening joint pain. She has had diffuse, aching pain in her knees, shoulders, and hands bilaterally for the past few months, but the pain has become much more severe in the past few weeks. She also reports night sweats and generalized malaise. On physical examination, radial and pedal pulses are weak. There are erythematous nodules over the legs that measure 3–5 cm. Laboratory studies show: Hematocrit 33.2% Hemoglobin 10.7 g/dL Leukocyte count 11,300/mm³ Platelet count 615,000/mm³ Erythrocyte sedimentation rate 94 mm/h Serum C-reactive protein 40 mg/dL (N=0.08–3.1) Which of the following is the most likely diagnosis?"

Answer Choices:

A. Temporal arteritis B. Polyarteritis nodosa C. Thromboangiitis obliterans D. Takayasu arteritis

Correct Answer: D. Takayasu arteritis

Question 114

Clinical Scenario:

A 20-year-old female presents to your clinic for evaluation. She complains of months of daily rhinorrhea, which she describes as watery and clear, as well as nasal congestion bilaterally. In addition, she reports frequent watery and itchy eyes, as well as daily sneezing. Her temperature is 100.1 deg F (37.8 deg C), blood pressure is 120/70 mmHg, pulse is 70/min, and respirations are 15/min. On exam, you note edematous, boggy turbinates with watery rhinorrhea. Which of the following is a treatment for the patient's condition?

Answer Choices:

A. Intravenous penicillin B. Oral amoxicillin C. Oral acetaminophen D. Intranasal fluticasone

Correct Answer: D. Intranasal fluticasone

Question 115

Clinical Scenario:

A 78-year-old man is brought to the emergency department by ambulance 30 minutes after the sudden onset of speech difficulties and right-sided arm and leg weakness. Examination shows paralysis and hypoesthesia on the right side, positive Babinski sign on the right, and slurred speech. A CT scan of the head shows a hyperdensity in the left middle cerebral artery and no evidence of intracranial bleeding. The patient's symptoms improve rapidly after pharmacotherapy is initiated and his weakness completely resolves. Which of the following drugs was most likely administered?

Answer Choices:

A. Rivaroxaban B. Alteplase C. Heparin D. Warfarin "

Correct Answer: B. Alteplase

Question 116

Clinical Scenario:

A 71-year old man is brought to the emergency department because of progressively worsening shortness of breath and fatigue for 3 days. During the last month, he has also noticed dark colored urine. He had an upper respiratory infection 6 weeks ago. He underwent a cholecystectomy at the age of 30 years. He has hypertension, hyperlipidemia, and type 2 diabetes mellitus. He immigrated to the US from Italy 50 years ago. Current medications include simvastatin, lisinopril, and metformin. He appears pale. His temperature is 37.1°C (98.8°F), pulse is 96/min, respirations are 21/min, and blood pressure is 150/80 mm Hg. Auscultation of the heart shows a grade 4/6 systolic murmur over the right second intercostal space that radiates to the carotids. Laboratory studies show: Leukocyte count 9,000/mm³ Hemoglobin 8.3 g/dL Hematocrit 24% Platelet count 180,000/mm³ LDH 212 U/L Haptoglobin 15 mg/dL (N=41–165) Serum Na⁺ 138 mEq/L K⁺ 4.5 mEq/L CL⁻ 102 mEq/L HCO₃⁻ 24 mEq/L Urea nitrogen 20 mg/dL Creatinine 1.2 mg/dL Total bilirubin 1.8 mg/dL Stool testing for occult blood is negative. Direct Coombs test is negative. Echocardiography shows an aortic jet velocity of 4.2 m/s and a mean pressure gradient of 46 mm Hg. Which of the following is the most appropriate next step in management to treat this patient's anemia?"

Answer Choices:

A. Administration of hydroxyurea B. Supplementation with iron C. Aortic valve replacement D. Discontinuation of medication "

Correct Answer: C. Aortic valve replacement

Question 117

Clinical Scenario:

A 75-year-old man is referred for thyroidectomy for treatment of thyroid nodules. A portion of the thyroid gland is resected, and the neck is closed with sutures. After awakening from anesthesia, the patient complains of 'hoarseness'. His vital signs are normal and his incisional pain is minimal. The surgeon realizes he most likely damaged the recurrent laryngeal nerve. Which of the following should the surgeon tell the patient?

Answer Choices:

A. "A mistake occurred during the operation, but there was no harm to you." B. "The case took longer than we thought it would, but everything went well." C. "The operation was successful and no complications occurred." D. "We made a mistake during the operation that may have harmed you."

Correct Answer: D. "We made a mistake during the operation that may have harmed you."

Question 118

Clinical Scenario:

A 50-year-old man presents to the emergency department with pain and swelling of his right leg for the past 2 days. Three days ago he collapsed on his leg after tripping on a rug. It was a hard fall and left him with bruising of his leg. Since then the pain and swelling of his leg have been gradually increasing. Past medical history is noncontributory. He lives a rather sedentary life and smokes two packs of cigarettes per day. The vital signs include heart rate 98/min, respiratory rate 15/min, temperature 37.8°C (100.1°F), and blood pressure 100/60 mm Hg. On physical examination, his right leg is visibly swollen up to the mid-calf with pitting edema and moderate erythema. Peripheral pulses in the right leg are weak and the leg is tender. Manipulation of the right leg is negative for Homan's sign. What is the next best step in the management of this patient?

Answer Choices:

A. Make a diagnosis of deep vein thrombosis based on history and physical B. Perform a venous ultrasound C. Start intravenous heparin therapy immediately D. Perform intravenous venography within 24 hours

Correct Answer: B. Perform a venous ultrasound

Question 119

Clinical Scenario:

Fifteen years ago, a physician was exposed to *Mycobacterium tuberculosis* during a medical mission trip to Haiti. A current CT scan of his chest reveals respiratory apical granulomas. The formation of this granuloma helped prevent the spread of the infection to other sites. Which pair of cells contributed to the walling-off of this infection?

Answer Choices:

A. TH1 cells and macrophages B. TH2 cells and macrophages C. TH1 cells and neutrophils D. CD8 T cells and NK cells

Correct Answer: A. TH1 cells and macrophages

Question 120

Clinical Scenario:

A 23-year-old woman with a past medical history significant for cardiac palpitations and hypothyroidism presents with cyclical lower abdominal pain and pelvic pain. Upon further questioning, she endorses difficulty conceiving over the last 12 months. On a review of systems, she endorses occasional pain with intercourse, which has become more frequent over the last 6 months. On physical examination, her heart and lungs are clear to auscultation, her abdomen has mild tenderness in the lower quadrants, and she shows normal range of motion in her extremities. Given the patient's desire to conceive, what is the most definitive treatment for her presumed condition?

Answer Choices:

A. NSAIDS B. Leuprolide C. Total abdominal hysterectomy with bilateral salpingo-oophorectomy (TAH-BSO)
D. Laparoscopy and lesion ablation

Correct Answer: D. Laparoscopy and lesion ablation

Question 121**Clinical Scenario:**

A 34-year-old woman comes to the physician for a follow-up appointment because of a blood pressure of 148/98 mm Hg at her last health maintenance examination four weeks ago. She feels well. She has a 20-year history of migraine with aura of moderate to severe intensity. For the past year, the headaches have been occurring 1–2 times per week. Her only medication is sumatriptan. She runs two to three times a week and does yoga once a week. She is sexually active with her husband and uses condoms inconsistently. Her father has type 2 diabetes mellitus and hypertension. Her temperature is 37.2°C (99.0°F), pulse is 76/min, respirations are 12/min, and blood pressure is 143/92 mm Hg. A repeat sitting blood pressure 20 minutes later is 145/94 mm Hg. Physical examination is unremarkable. Which of the following is the most appropriate pharmacotherapy for this patient?

Answer Choices:

A. Hydrochlorothiazide B. Lisinopril C. Propranolol D. Prazosin

Correct Answer: C. Propranolol

Question 122**Clinical Scenario:**

A 46-year-old man is brought to the emergency department because of severe epigastric pain and vomiting for the past 4 hours. The pain is constant, radiates to his back, and is worse on lying down. He has had 3–4 episodes of greenish-colored vomit. He was treated for *H. pylori* infection around 2 months ago with triple-regimen therapy. He has atrial fibrillation and hypertension. He owns a distillery on the outskirts of a town. The patient drinks 4–5 alcoholic beverages daily. Current medications include dabigatran and metoprolol. He appears uncomfortable. His temperature is 37.8°C (100°F), pulse is 102/min, and blood pressure is 138/86 mm Hg. Examination shows severe epigastric tenderness to palpation with guarding but no rebound. Bowel sounds are hypoactive. Rectal examination shows no abnormalities. Laboratory studies show: Hematocrit 53% Leukocyte count 11,300/mm³ Serum Na⁺ 133 mEq/L Cl⁻ 98 mEq/L K⁺ 3.1 mEq/L Calcium 7.8 mg/dL Urea nitrogen 43 mg/dL Glucose 271 mg/dL Creatinine 2.0 mg/dL Total bilirubin 0.7 mg/dL Alkaline phosphatase 61 U/L AST 19 U/L ALT 17 U/L γ -glutamyl transferase (GGT) 88 u/L (N=5–50 U/L) Lipase 900 U/L (N=14–280 U/L) Which of the following is the most appropriate next step in management?"

Answer Choices:

A. Calcium gluconate therapy B. Fomepizole therapy C. Laparotomy D. Crystalloid fluid infusion "

Correct Answer: D. Crystalloid fluid infusion "

Question 123

Clinical Scenario:

A 28-year-old woman comes to the emergency department because of increasing weakness and numbness of her legs for 3 days. She noticed that the weakness was more severe after she had a hot shower that morning. A year ago, she had an episode of partial vision loss in her left eye that resolved within 3 weeks. She is sexually active with 3 male partners and uses condoms inconsistently. She appears anxious. Her temperature is 37°C (98.6°F), pulse is 80/min, and blood pressure is 108/77 mm Hg. Examination shows spasticity and decreased muscle strength in bilateral lower extremities. Deep tendon reflexes are 4+ bilaterally. Plantar reflex shows an extensor response bilaterally. The abdominal reflex is absent. Sensation to vibration and position over the lower extremities shows no abnormalities. Tandem gait is impaired. MRI of the brain and spine is inconclusive. Further evaluation is most likely to show which of the following?

Answer Choices:

A. Positive rapid plasma reagin test B. Slow nerve conduction velocity C. Elevated intrinsic factor antibody level D. Oligoclonal bands in cerebral spinal fluid

Correct Answer: D. Oligoclonal bands in cerebral spinal fluid

Question 124

Clinical Scenario:

A 35-year-old woman comes to the physician because of a 1-day history of swelling and pain in the left leg. Two days ago, she returned from a business trip on a long-distance flight. She has alcohol use disorder. Physical examination shows a tender, swollen, and warm left calf. Serum studies show an increased homocysteine concentration and a methylmalonic acid concentration within the reference range. Further evaluation of this patient is most likely to show which of the following serum findings?

Answer Choices:

A. Increased pyridoxine concentration B. Increased fibrinogen concentration C. Decreased cobalamin concentration D. Decreased folate concentration

Correct Answer: D. Decreased folate concentration

Question 125

Clinical Scenario:

A 35-year-old man presents with a mass on the central part of his neck. He reports it has been growing steadily for the past 2 weeks, and he has also been experiencing fatigue and recurrent fevers. No significant past

medical history. The patient denies any smoking history, or alcohol or recreational drug use. He denies any recent travel in the previous 6 months. On physical examination, there are multiple enlarged submandibular and cervical lymph nodes that are firm, mobile, and non-tender. A biopsy of one of the lymph nodes is performed and shows predominantly lymphocytes and histiocytes present in a pattern 'resembling popcorn'. A flow cytometry analysis demonstrates cells that are CD19 and CD20 positive and CD15 and CD30 negative. Which of the following is the most likely diagnosis in this patient?

Answer Choices:

A. Lymphocyte rich classical Hodgkin lymphoma B. Nodular lymphocyte-predominant Hodgkin lymphoma C. Nodular sclerosis classical Hodgkin lymphoma D. Lymphocyte depleted Hodgkin lymphoma

Correct Answer: B. Nodular lymphocyte-predominant Hodgkin lymphoma

Question 126

Clinical Scenario:

A 25-year-old pregnant woman at 28 weeks gestation presents with a headache. Her pregnancy has been managed by a nurse practitioner. Her temperature is 99.0°F (37.2°C), blood pressure is 164/104 mmHg, pulse is 100/min, respirations are 22/min, and oxygen saturation is 98% on room air. Physical exam is notable for a comfortable appearing woman with a gravid uterus. Laboratory tests are ordered as seen below.

Hemoglobin: 12 g/dL Hematocrit: 36% Leukocyte count: 6,700/mm³ with normal differential Platelet count: 100,500/mm³

Serum: Na⁺: 141 mEq/L Cl⁻: 101 mEq/L K⁺: 4.4 mEq/L HCO₃⁻: 25 mEq/L BUN: 21 mg/dL Glucose: 99 mg/dL Creatinine: 1.0 mg/dL AST: 32 U/L ALT: 30 U/L

Urine: Color: Amber Protein: Positive Blood: Negative

Which of the following is the most likely diagnosis?

Answer Choices:

A. Acute fatty liver disease of pregnancy B. Eclampsia C. Preeclampsia D. Severe preeclampsia

Correct Answer: D. Severe preeclampsia

Question 127

Clinical Scenario:

A 68-year-old man presents for a screening ultrasound scan. He has been feeling well and is in his usual state of good health. His medical history is notable for mild hypertension and a 100-pack-year tobacco history. He has a blood pressure of 128/86 and heart rate of 62/min. Physical examination is clear lung sounds and regular heart sounds. On ultrasound, an infrarenal aortic aneurysm of 4 cm in diameter is identified. Which of the following is the best initial step for this patient?

Answer Choices:

A. Beta-blockers B. Surveillance C. Urgent repair D. Reassurance

Correct Answer: B. Surveillance

Question 128

Clinical Scenario:

A 25-year-old woman presents with a history of recurrent attacks of unprovoked fear, palpitations, and fainting. The attacks are usually triggered by entering a crowded place or public transport, so the patient tries to avoid being in public places alone. Besides this, she complains of difficulties in falling asleep, uncontrolled worry about her job and health, fear to lose the trust of her friends, and poor appetite. She enjoys dancing and has not lost a passion for her hobby, but recently when she participated in a local competition, she had an attack which made her stop her performance until she calmed down and her condition improved. She feels upset due to her condition. She works as a sales manager and describes her work as demanding with multiple deadlines to be met. She recently broke up with her boyfriend. She does not report any chronic medical problems, but she sometimes takes doxylamine to fall asleep. She has a 4-pack-year history of smoking and drinks alcohol occasionally. On presentation, her blood pressure is 110/60 mm Hg, heart rate is 71/min, respiratory rate is 13/min, and temperature is 36.5°C (97.7°F). Her physical examination is unremarkable. Which of the following medications can be used for the acute management of the patient's attacks?

Answer Choices:

A. Bupropion B. Metoprolol C. Clonazepam D. Nifedipine

Correct Answer: C. Clonazepam

Question 129

Clinical Scenario:

A 23-year-old woman comes to the physician because of increased urinary frequency and pain on urination for two days. She has had three similar episodes over the past year that resolved with antibiotic treatment. She has no history of serious illness. She is sexually active with one male partner; they do not use barrier contraception. Upon questioning, she reports that she always urinates and cleans herself after sexual intercourse. She drinks 2–3 liters of fluid daily. Her only medication is a combined oral contraceptive. Her temperature is 36.9°C (98.4°F), pulse is 65/min, and blood pressure is 122/65 mm Hg. Examination shows mild tenderness to palpation in the lower abdomen. The remainder of the examination shows no abnormalities. Urinalysis shows WBCs and rare gram-positive cocci. Which of the following is the most appropriate recommendation to prevent similar episodes in the future?

Answer Choices:

A. Postcoital vaginal probiotics B. Treatment of the partner with intramuscular ceftriaxone C. Postcoital oral amoxicillin-clavulanate D. Daily oral trimethoprim-sulfamethoxazole "

Correct Answer: D. Daily oral trimethoprim-sulfamethoxazole "

Question 130

Clinical Scenario:

A 56-year-old postmenopausal woman comes to the physician because of a 6-month history of worsening pain and swelling in her left knee. She has a history of peptic ulcer disease for which she takes cimetidine. Examination shows palpable crepitus and limited range of motion of the left knee. Which of the following is the most appropriate pharmacotherapy for this patient's symptoms?

Answer Choices:

A. Ketorolac B. Diclofenac C. Acetylsalicylic acid D. Celecoxib

Correct Answer: D. Celecoxib

Question 131

Clinical Scenario:

A 30-year-old man presents with fatigue and low energy. He says that he has been "feeling down" and tired on most days for the last 3 years. He also says that he has had difficulty concentrating and has been sleeping excessively. The patient denies any manic or hypomanic symptoms. He also denies any suicidal ideation or preoccupation with death. A physical examination is unremarkable. Laboratory findings are significant for the following: Serum glucose (fasting) 88 mg/dL Serum electrolytes Sodium 142 mEq/L; Potassium: 3.9 mEq/L; Chloride: 101 mEq/L Serum creatinine 0.8 mg/dL Blood urea nitrogen 10 mg/dL Hemoglobin (Hb %) 15 g/dL Mean corpuscular volume (MCV) 85 fl Reticulocyte count 1% Erythrocyte count 5.1 million/mm³ Thyroid-stimulating hormone 3.5 μ U/mL Medication is prescribed to this patient that increases norepinephrine nerve stimulation. After 2 weeks, the patient returns for follow-up and complains of dizziness, dry mouth, and constipation. Which of the following drugs was most likely prescribed to this patient?

Answer Choices:

A. Clonidine B. Venlafaxine C. Lithium D. Phenylephrine

Correct Answer: B. Venlafaxine

Question 132

Clinical Scenario:

A scientist is studying the process of thymus-dependent B cell activation in humans. He observes that, after bacterial infections, the germinal centers of secondary lymphoid organs become highly metabolically active. After subsequent reinfection with the same pathogen, the organism is able to produce immunoglobulins at a much faster pace. Which of the following processes is likely taking place in the germinal centers at the beginning of an infection?

Answer Choices:

A. Development of early pro-B cells B. Development of immature B cells C. T cell negative selection D. Affinity maturation

Correct Answer: D. Affinity maturation

Question 133

Clinical Scenario:

A grant reviewer at the National Institutes of Health is determining which of two studies investigating the effects of gastric bypass surgery on fasting blood sugar to fund. Study A is spearheaded by a world renowned surgeon, is a multi-center study planning to enroll 50 patients at each of 5 different sites, and is single-blinded. Study B plans to enroll 300 patients from a single site and will be double-blinded by virtue of a sham surgery for the control group. The studies both plan to use a t-test, and they both report identical expected treatment effect sizes and variance. If the reviewer were interested only in which trial has the higher power, which proposal should he fund?

Answer Choices:

A. Study A, because it has a superior surgeon B. Study A, because it is a multi-center trial C. Study B, because it has a larger sample size D. Both studies have the same power

Correct Answer: C. Study B, because it has a larger sample size

Question 134

Clinical Scenario:

A 25-year-old woman is brought to the emergency department because of a 1-day history of lower abdominal pain and vaginal bleeding. Her last menstrual period was 7 weeks ago. A urine pregnancy test is positive. A pelvic ultrasound shows a normal appearing uterus with an empty intrauterine cavity and a minimal amount of free pelvic fluid. Treatment with a drug is begun. Which of the following is the most likely effect of this drug?

Answer Choices:

A. Decrease in guanylate B. Increase in deoxyuridine monophosphate C. Decrease in phosphoribosyl pyrophosphate D. Increase in tetrahydrofolate polyglutamate

Correct Answer: B. Increase in deoxyuridine monophosphate

Question 135

Clinical Scenario:

A 10-day-old male infant is brought to the emergency room for abdominal distension for the past day. His mother reports that he has been refusing feeds for about 1 day and appears more lethargic than usual. While changing his diaper today, she noticed that the baby felt warm. He has about 1-2 wet diapers a day and has 1-2

seedy stools a day. The mother reports an uncomplicated vaginal delivery. His past medical history is significant for moderate respiratory distress following birth that has since resolved. His temperature is 101°F (38.3°C), blood pressure is 98/69 mmHg, pulse is 174/min, respirations are 47/min, and oxygen saturation is 99% on room air. A physical examination demonstrates a baby in moderate distress with abdominal distension. What is the best initial step in the management of this patient?

Answer Choices:

A. Cystoscopy B. Radionuclide scan C. Renal ultrasound D. Urinary catheterization

Correct Answer: D. Urinary catheterization

Question 136

Clinical Scenario:

A 41-year-old woman comes to the physician because of an 8-hour history of colicky abdominal pain and nausea. The pain worsened after she ate a sandwich, and she has vomited once. She has no history of serious medical illness. Her temperature is 37.2°C (99.1°F), pulse is 80/min, and blood pressure is 134/83 mm Hg. Physical examination shows scleral icterus and diffuse tenderness in the upper abdomen. Serum studies show: Total bilirubin 2.7 mg/dL AST 35 U/L ALT 38 U/L Alkaline phosphatase 180 U/L γ -Glutamyltransferase 90 U/L (N = 5–50) Ultrasonography is most likely to show a stone located in which of the following structures?"

Answer Choices:

A. Common bile duct B. Common hepatic duct C. Cystic duct D. Gallbladder fundus

Correct Answer: A. Common bile duct

Question 137

Clinical Scenario:

A 62-year-old woman is brought to the physician by her daughter for the evaluation of weight loss and a bloody cough that began 3 weeks ago. Twenty years ago, she had a major depressive episode and a suicide attempt. Since then, her mental status has been stable. She lives alone and takes care of all her activities of daily living. The patient has smoked 1 pack of cigarettes daily for the past 40 years. She does not take any medications. An x-ray of the chest shows a central solitary nodule in the right lung; bronchoscopy with transbronchial biopsy shows a small cell lung cancer. A CT scan of the abdomen shows multiple metastatic lesions within the liver. The patient previously designated her daughter as her healthcare decision-maker. As the physician goes to reveal the diagnosis to the patient, the patient's daughter is waiting outside her room. The daughter asks the physician not to tell her mother the diagnosis. Which of the following is the most appropriate action by the physician?

Answer Choices:

A. Ask the patient if she wants to know the truth B. Disclose the diagnosis to the patient C. Encourage the daughter to disclose the diagnosis to her mother D. Clarify the daughter's reasons for the request

Correct Answer: D. Clarify the daughter's reasons for the request

Question 138

Clinical Scenario:

A 28-year-old man is admitted to the hospital for the evaluation of symmetric, ascending weakness that started in his feet and has become progressively worse over the past 5 days. A lumbar puncture is performed to confirm the diagnosis. As the needle is advanced during the procedure, there is resistance just before entering the epidural space. This resistance is most likely due to which of the following structures?

Answer Choices:

A. Ligamentum flavum B. Superficial fascia C. Interspinous ligament D. Supraspinous ligament

Correct Answer: A. Ligamentum flavum

Question 139

Clinical Scenario:

A 44-year-old woman comes to the physician for the evaluation of a 1-month history of fatigue and difficulty swallowing. During this period, she has also had dry skin, thinning hair, and rounding of her face. She has type 1 diabetes mellitus and rheumatoid arthritis. Her father had a thyroidectomy for papillary thyroid cancer. The patient had smoked one pack of cigarettes daily for 20 years but quit 3 years ago. She drinks 2–3 glasses of wine daily. Her current medications include insulin, omeprazole, and daily ibuprofen. She appears well. Her temperature is 36.3°C (97.3°F), pulse is 62/min, and blood pressure is 102/76 mm Hg. Examination of the neck shows a painless, diffusely enlarged thyroid gland. Cardiopulmonary examination shows no abnormalities. Further evaluation is most likely to show which of the following?

Answer Choices:

A. Increased uptake on radioactive iodine scan in discrete 1-cm area B. Diffusely increased uptake on a radioactive iodine scan C. Positive immunohistochemical stain for calcitonin on thyroid biopsy D. Positive thyroid peroxidase antibodies and thyroglobulin antibodies in serum

Correct Answer: D. Positive thyroid peroxidase antibodies and thyroglobulin antibodies in serum

Question 140

Clinical Scenario:

A 25-year-old woman with bipolar disorder and schizophrenia presents to the emergency room stating that she is pregnant. She says that she has been pregnant since she was 20 years old and is expecting a baby now that she is breathing much harder and feeling more faint with chest pain caused by deep breaths. Her hospital medical record shows multiple negative pregnancy tests over the past 5 years. The patient has a 20 pack-year

smoking history. Her temperature is 98°F (37°C), blood pressure is 100/60 mmHg, pulse is 110/min, respirations are 28/min, and oxygen saturation is 90% on room air. Her fingerstick glucose is 100 mg/dL. She has a large abdominal pannus which is soft and nontender. Her legs are symmetric and non-tender. Oxygen is provided via nasal cannula. Her urine pregnancy test comes back positive and an initial chest radiograph is unremarkable. What is the next best step in diagnosis?

Answer Choices:

A. CT angiogram B. D-dimer C. Ultrasound D. Ventilation-perfusion scan

Correct Answer: D. Ventilation-perfusion scan

Question 141

Clinical Scenario:

A 28-year-old man comes to the physician because of skin lesions on and around his anus. He noticed them 3 days ago. The lesions are not painful and he does not have any urinary complaints. He has smoked one pack of cigarettes daily for 10 years and he drinks 6–7 beers on weekends. He is sexually active with two male partners and uses condoms inconsistently. He appears healthy. A photograph of the perianal region is shown. The lesions turn white after application of a dilute acetic acid solution. The remainder of the examination shows no abnormalities. An HIV test is negative. Which of the following is the most appropriate next step in management?

Answer Choices:

A. Oral acyclovir B. Topical mometasone C. Parenteral benzathine penicillin D. Curettage

Correct Answer: D. Curettage

Question 142

Clinical Scenario:

A 44-year-old man comes to the physician for a pre-employment evaluation. On questioning, he reports a mild cough, sore throat, and occasional headaches for 1 week. He has not had fever or weight loss. Nine years ago, he was diagnosed with HIV. He has gastroesophageal reflux disease. He has a history of IV drug abuse but quit 8 years ago. He has smoked one pack of cigarettes daily for 27 years and does not drink alcohol. Current medications include tenofovir, emtricitabine, efavirenz, and esomeprazole. He is 180 cm (5 ft 11 in) tall and weighs 89 kg (196 lbs); BMI is 27.5 kg/m². His temperature is 37.3°C (99.1°F), pulse is 81/min, respirations are 17/min, and blood pressure is 145/75 mm Hg. Pulmonary examination shows no abnormalities. There are a few scattered old scars along the left elbow flexure. Laboratory studies show a leukocyte count of 6200/mm³, hemoglobin of 13.8 g/dL, and CD4⁺ count of 700/mm³ (N = ≥ 500/mm³). A tuberculin skin test (TST) comes back after 50 hours with an induration of 3 mm in diameter. Which of the following is the most appropriate next step in management?

Answer Choices:

A. Chest x-ray B. Reassurance C. Interferon- γ release assay D. Repeat tuberculin skin test after 6–8 weeks

Correct Answer: B. Reassurance

Question 143

Clinical Scenario:

A 59-year-old man presents with intense, sharp pain in his toe for the past hour. He reports similar symptoms in the past and this is his 2nd visit to the emergency department this year with the same complaint. The patient is afebrile and the vital signs are within normal limits. On physical examination, there is significant erythema, swelling, warmth, and moderate pain on palpation of the right 1st toe. The remainder of the examination is unremarkable. A plain radiograph of the right foot reveals no abnormalities. Joint arthrocentesis of the inflamed toe reveals urate crystals. Laboratory studies show: Serum glucose (random) 170 mg/dL Sodium 140 mEq/L Potassium 4.1 mEq/L Chloride 100 mEq/L Uric acid 7.2 mg/dL Serum creatinine 0.8 mg/dL Blood urea nitrogen 9 mg/dL Cholesterol, total 170 mg/dL HDL-cholesterol 43 mg/dL LDL-cholesterol 73 mg/dL Triglycerides 135 mg/dL HDL: high-density lipoprotein; LDL: low-density lipoprotein Ibuprofen is prescribed for the acute treatment of this patient's symptoms. He is also put on chronic therapy to prevent the recurrence of future attacks. Which of the following drugs is 1st-line for chronic therapy of gout?

Answer Choices:

A. Methotrexate B. Colchicine C. Allopurinol D. Indomethacin

Correct Answer: C. Allopurinol

Question 144

Clinical Scenario:

A 45-year-old man is brought to the physician for a follow-up examination. Three weeks ago, he was hospitalized and treated for spontaneous bacterial peritonitis. He has alcoholic liver cirrhosis and hypothyroidism. His current medications include spironolactone, lactulose, levothyroxine, trimethoprim-sulfamethoxazole, and furosemide. He appears ill. His temperature is 36.8°C (98.2°F), pulse is 77/min, and blood pressure is 106/68 mm Hg. He is oriented to place and person only. Examination shows scleral icterus and jaundice. There is 3+ pedal edema and reddening of the palms bilaterally. Breast tissue appears enlarged, and several telangiectasias are visible over the chest and back. Abdominal examination shows dilated tortuous veins. On percussion of the abdomen, the fluid-air level shifts when the patient moves from lying supine to right lateral decubitus. Breath sounds are decreased over both lung bases. Cardiac examination shows no abnormalities. Bilateral tremor is seen when the wrists are extended. Genital examination shows reduced testicular volume of both testes. Digital rectal examination and proctoscopy show hemorrhoids. Which of the following potential complications of this patient's condition is the best indication for the placement of a transjugular intrahepatic portosystemic shunt (TIPS)?

Answer Choices:

A. Hepatic veno-occlusive disease B. Recurrent variceal hemorrhage C. Portal hypertensive gastropathy D. Hepatic hydrothorax

Correct Answer: B. Recurrent variceal hemorrhage

Question 145

Clinical Scenario:

A 45-year-old male is presenting for routine health maintenance. He has no complaints. His pulse is 75/min, blood pressure is 155/90 mm Hg, and respiratory rate is 15/min. His body mass index is 25 kg/m². The physical exam is within normal limits. He denies any shortness of breath, daytime sleepiness, headaches, sweating, or palpitations. He does not recall having an elevated blood pressure measurement before. Which of the following is the best next step?

Answer Choices:

A. Refer patient to cardiologist B. Treat with thiazide diuretic C. Repeat the blood pressure measurement D. Provide reassurance

Correct Answer: C. Repeat the blood pressure measurement

Question 146

Clinical Scenario:

A 770-g (1-lb 11-oz) female newborn delivered at 28 weeks' gestation develops rapid breathing, grunting, cyanosis, and subcostal retractions shortly after birth. Her mother did not receive any prenatal care. Breath sounds are decreased over both lung fields. An x-ray of the chest shows diffuse fine, reticular densities bilaterally. Antenatal administration of which of the following drugs would most likely have prevented this infant's current condition?

Answer Choices:

A. Epinephrine B. Betamethasone C. Thyrotropin-releasing hormone D. Oxytocin

Correct Answer: B. Betamethasone

Question 147

Clinical Scenario:

A 26-year-old man is brought to the emergency room by his roommate after he was found attempting to commit suicide. His roommate says that he stopped him before he was about to jump off the balcony. He has been receiving treatment for depression for about a year. 6 months ago, he had come to the hospital reporting decreased interest in his daily activities and inability to concentrate on his work. He had stopped going out or accepting invitations for any social events. He spent several nights tossing and turning in bed. He also

expressed guilt for being unable to live up to his parents' expectations. His psychiatrist started him on fluoxetine. He says that none of the medications have helped even though the dose of his medication was increased on several occasions, and he was also switched to other medications over the course of the past year. He has mentioned having suicidal thoughts due to his inability to cope with daily activities, but this is the first time he has ever attempted it. Which of the following would this patient be a suitable candidate for?

Answer Choices:

A. Cognitive behavioral theory B. Electroconvulsive therapy C. Amitriptyline D. Olanzapine

Correct Answer: B. Electroconvulsive therapy

Question 148

Clinical Scenario:

A 31-year-old physician notices that her senior colleague has been arriving late for work for the past 2 weeks. The colleague recently lost his wife to cancer and has been taking care of his four young children. Following the death of his wife, the department chair offered him extended time off but he declined. There have been some recent changes noted in this colleague that have been discussed among the resident physicians, such as missed clinic appointments, two intra-operative errors, and the smell of alcohol on his breath on three different occasions. Which of the following is the most appropriate action by the physician regarding her colleague?

Answer Choices:

A. Confront the colleague in private B. Inform the local Physician Health Program C. Inform the colleague's patients about the potential hazard D. Alert the State Licensing Board

Correct Answer: B. Inform the local Physician Health Program

Question 149

Clinical Scenario:

A 65-year-old man comes to the physician because of fatigue and nausea for 1 week. Over the past six months, he has had to get up twice every night to urinate. Occasionally, he has had discomfort during urination. He has arterial hypertension. His father died of renal cell carcinoma. Current medications include ramipril. His temperature is 37.3°C (99.1°F), pulse is 88/min, and blood pressure is 124/78 mm Hg. The abdomen is soft and nontender. Cardiac and pulmonary examinations show no abnormalities. Rectal examination shows a symmetrically enlarged and smooth prostate. Serum studies show: Hemoglobin 14.9 g/dL Leukocyte count 7500/mm³ Platelet count 215,000/mm³ Serum Na⁺ 136 mEq/L Cl⁻ 101 mEq/L K⁺ 4.9 mEq/L HCO₃⁻ 23 mEq/L Glucose 95 mg/dL Urea nitrogen 25 mg/dL Creatinine 1.9 mg/dL PSA 2.1 ng/mL (normal <4 ng/mL) Urine Blood negative Protein 1+ Glucose negative RBC casts negative Which of the following is the most appropriate next step in management?"

Answer Choices:

A. CT scan of the abdomen and pelvis B. Transrectal ultrasonography C. Renal ultrasonography D. Ureteral stenting

Correct Answer: C. Renal ultrasonography

Question 150

Clinical Scenario:

A 24-year-old African American college student comes to the office for a scheduled visit. He has been healthy, although he reports occasional flank discomfort which comes and goes. He denies any fever, chills, dysuria, or polyuria in the past year. His vaccinations are up to date. His family history is unknown, as he was adopted. He smokes 1 pack of cigarettes every 3 days, drinks socially, and denies any current illicit drug use, although he endorses a history of injection drug use. He currently works as a waiter to afford his college tuition. His physical examination shows a young man with a lean build, normal heart sounds, clear breath sounds, bowel sounds within normal limits, and no lower extremity edema. You order a urinalysis which shows 8 red blood cells (RBCs) per high-power field (HPF). The test is repeated several weeks later and shows 6 RBCs/HPF. What is the most appropriate next step in management?

Answer Choices:

A. Plain abdominal X-ray B. Intravenous (IV) pyelogram C. 24-hour urine collection test D. Repeat urinalysis in 6 months

Correct Answer: B. Intravenous (IV) pyelogram

Question 151

Clinical Scenario:

A 68-year-old man with hypertension comes to the physician because of fatigue and difficulty initiating urination. He wakes up several times a night to urinate. He does not take any medications. His blood pressure is 166/82 mm Hg. Digital rectal examination shows a firm, non-tender, and uniformly enlarged prostate. Which of the following is the most appropriate pharmacotherapy?

Answer Choices:

A. Phenoxybenzamine B. Tamsulosin C. Terazosin D. α -Methyldopa

Correct Answer: C. Terazosin

Question 152

Clinical Scenario:

A 70-year-old Caucasian male visits your office regularly for treatment of New York Heart association class IV congestive heart failure. Which of the following medications would you add to this man's drug regimen in order to improve his overall survival?

Answer Choices:

A. Spironolactone B. Amiloride C. Hydrochlorothiazide D. Acetazolamide

Correct Answer: A. Spironolactone

Question 153

Clinical Scenario:

A 21-year-old man presents to the office for a follow-up visit. He was recently diagnosed with type 1 diabetes mellitus after being hospitalized for diabetic ketoacidosis following a respiratory infection. He is here today to discuss treatment options available for his condition. The doctor mentions a recent study in which researchers have developed a new version of the insulin pump that appears efficacious in type 1 diabetics. They are currently comparing it to insulin injection therapy. This new pump is not yet available, but it looks very promising. At what stage of clinical trials is this current treatment most likely at?

Answer Choices:

A. Phase 4 B. Phase 3 C. Phase 0 D. Phase 2

Correct Answer: B. Phase 3

Question 154

Clinical Scenario:

A previously healthy 25-year-old man is brought to the emergency department 30 minutes after collapsing during soccer practice. His father died of sudden cardiac arrest at the age of 36 years. The patient appears well. His pulse is 73/min and blood pressure is 125/78 mm Hg. Cardiac examination is shown. An ECG shows large R waves in the lateral leads and deep S waves in V1 and V2. Further evaluation is most likely to show which of the following?

Answer Choices:

A. Aortic root dilatation B. Eccentric left ventricular dilation C. Asymmetric septal hypertrophy D. Mitral valve fibrinoid necrosis

Correct Answer: C. Asymmetric septal hypertrophy

Question 155

Clinical Scenario:

A 74-year-old man presents to the emergency department with shortness of breath that started about 30 mins ago. He is also experiencing chest pain on deep inspiration. He has several significant medical conditions including obstructive pulmonary disease, hypertension, and dyslipidemia. He used to smoke about 3 packs of cigarettes every day until last year when he quit. He was in the emergency room 2 weeks ago for a hemorrhagic stroke which was promptly treated. He currently weighs 97.5 kg (215 lb). His respirations are 20/min, the blood pressure is 110/ 80 mm Hg, and the pulse is 105/min. On physical examination, Homan's sign is positive. An ECG and chest X-ray are performed. His current oxygen saturation is at 87% and D-dimer results are positive. He is wheeled in for a CT scan. Which of the following treatments would this patient be an ideal candidate for?

Answer Choices:

A. Greenfield filter B. Unfractionated heparin C. Subcutaneous fondaparinux D. Warfarin

Correct Answer: A. Greenfield filter

Question 156

Clinical Scenario:

A 54-year-old man presents with feelings of sadness and low mood on most days of the week for the past month. He reports an inability to concentrate and also finds it hard to develop an interest in his daily activities. He goes on to tell his physician that he struggles to get sleep at night, and, in the morning, he doesn't have the energy to get out of bed. He says he feels like a loser since he hasn't accomplished all that he had set out to do and feels guilty for being unable to provide for his family. He says he doesn't have the will to live anymore but denies any plans to commit suicide. Past medical history is significant for erectile dysfunction which was diagnosed about a year ago. Which of the following medications should be avoided in the treatment of this patient's depression?

Answer Choices:

A. Vortioxetine B. Fluoxetine C. Bupropion D. Vilazodone

Correct Answer: B. Fluoxetine

Question 157

Clinical Scenario:

A 54-year-old woman comes to the clinic for an annual check-up. She has no other complaints except for some weight gain over the past year. Her last menstrual period was 8 months ago. "I started eating less since I get full easily and exercising more but just can't lose this belly fat," she complains. She is sexually active with her husband and does not use any contraception since "I am old." She denies vaginal dryness, hot flashes, fevers, abdominal pain, or abnormal vaginal bleeding but does endorse intermittent constipation for the past year. Physical examination is unremarkable except for some mild abdominal distension with fluid wave. Laboratory findings are as follows:

Serum: Na+: 138 mEq/L Cl-: 97 mEq/L K+: 3.9 mEq/L Urea nitrogen: 21 mg/dL Creatinine: 1.4 mg/dL
Glucose: 120 mg/dL B-hCG: negative

What is the most likely diagnosis for this patient?

Answer Choices:

A. Endometriosis B. Menopause C. Ovarian cancer D. Pregnancy

Correct Answer: C. Ovarian cancer

Question 158

Clinical Scenario:

A 26-year-old female with AIDS (CD4 count: 47) presents to the emergency department in severe pain. She states that over the past week she has been fatigued and has had a progressively worse headache and fever. These symptoms have failed to remit leading her to seek care in the ED. A lumbar puncture is performed which demonstrates an opening pressure of 285 mm H₂O, increased lymphocytes, elevated protein, and decreased glucose. The emergency physician subsequently initiates treatment with IV amphotericin B and PO flucytosine. What additional treatment in the acute setting may be warranted in this patient?

Answer Choices:

A. Fluconazole B. Serial lumbar punctures C. Mannitol D. Acetazolamide

Correct Answer: B. Serial lumbar punctures

Question 159

Clinical Scenario:

Your test subject is a stout 52-year-old gentleman participating in a study on digestion. After eating a platter of meat riblets and beef strips a test subjects digestive tract undergoes vast hormonal changes. Which of the following changes likely occurred in this patient as a result of the meal?

Answer Choices:

A. Increased gastrin release leading to a decrease in proton secretion B. Decreased cholecystokinin release from the I cells of the duodenum C. Increased release of secretin from S cells of the duodenum D. Decreased Ach release from the vagus nerve

Correct Answer: C. Increased release of secretin from S cells of the duodenum

Question 160

Clinical Scenario:

A 63-year-old retired teacher presents to his family physician for an annual visit. He has been healthy for most of his life and currently takes no medications, although he has had elevated blood pressure on several visits in the past few years but declined taking any medication. He has no complaints about his health and has been enjoying time with his grandchildren. He has been a smoker for 40 years—ranging from half to 1 pack a day, and he drinks 1 beer daily. On presentation, his blood pressure is 151/98 mm Hg in both arms, heart rate is 89/min, and respiratory rate is 14/min. Physical examination reveals a well-appearing man with no physical abnormalities. A urinalysis is performed and shows microscopic hematuria. Which of the following is the best next step for this patient?

Answer Choices:

A. Perform a CT scan of the abdomen with contrast B. Perform intravenous pyelography C. Perform a cystoscopy D. Repeat the urinalysis

Correct Answer: D. Repeat the urinalysis

Question 161

Clinical Scenario:

Background: Aldosterone blockade reduces mortality and morbidity among patients with severe heart failure. Researchers conducted a double-blind, placebo-controlled study evaluating the effect of eplerenone, a selective aldosterone blocker, on morbidity and mortality among patients with acute myocardial infarction complicated by left ventricular dysfunction and heart failure. Methods: Patients were randomly assigned to eplerenone (25 mg per day initially, titrated to a maximum of 50 mg per day; 3,319 patients) or placebo (3,313 patients) in addition to optimal medical therapy. The study continued until 1,012 deaths occurred. The primary endpoints were death from any cause, death from cardiovascular causes, hospitalization for heart failure, acute myocardial infarction, stroke, or ventricular arrhythmia. Results: During a mean follow-up of 16 months, there were 478 deaths in the eplerenone group (14.4%) and 554 deaths in the placebo group (16.7%, $p = 0.008$). Of these deaths, 407 in the eplerenone group and 483 in the placebo group were attributed to cardiovascular causes (relative risk, 0.83; 95 percent confidence interval, 0.72 to 0.94; $p = 0.005$). The rate of the other primary endpoints, death from cardiovascular causes or hospitalization for cardiovascular events, was reduced by eplerenone (relative risk, 0.87; 95 percent confidence interval, 0.79 to 0.95; $p = 0.002$), as was the secondary endpoint of death from any cause or any hospitalization (relative risk, 0.92; 95 percent confidence interval, 0.86 to 0.98; $p = 0.02$). There was also a reduction in the rate of sudden death from cardiac causes (relative risk, 0.79; 95 percent confidence interval, 0.64 to 0.97; $p = 0.03$). The rate of serious hyperkalemia was 5.5 percent in the eplerenone group and 3.9 percent in the placebo group ($p = 0.002$), whereas the rate of hypokalemia was 8.4 percent in the eplerenone group and 13.1 percent in the placebo group ($p < 0.001$). Which of the following represents the number of patients needed to treat to save one life, based on the primary endpoint?

Answer Choices:

A. $1/(0.136 - 0.118)$ B. $1/(0.300 - 0.267)$ C. $1/(0.167 - 0.144)$ D. $1/(0.267 - 0.300)$

Correct Answer: C. $1/(0.167 - 0.144)$

Question 162

Clinical Scenario:

A 60-year-old man comes to the physician because of recurrent nose bleeds that occur with light trauma or at random times during the day. Over the past 6 months, the patient has felt weak and fatigued and has had a 10-kg (22-lb) weight loss. He has poor appetite and describes abdominal discomfort. He does not have night sweats. His pulse is 72/min, blood pressure is 130/70 mm Hg, and his temperature is 37.5°C (99.5°F). The spleen is palpated 10 cm below the left costal margin. Multiple bruises are noted on both upper extremities. Laboratory studies show. Hemoglobin 9.8 g/dL Hematocrit 29.9% Leukocyte count 4,500/mm³ Neutrophils 30% Platelet count 74,000/mm³ Serum Lactate dehydrogenase 410 IU/L A peripheral blood smear detects tartrate-resistant acid phosphatase activity. Which of the following is the most appropriate initial treatment for this patient?"

Answer Choices:

A. Transfusion of packed red blood cells B. Transfusion of platelets C. Melphalan D. Cladribine

Correct Answer: D. Cladribine

Question 163

Clinical Scenario:

A 7-year-old girl is brought to the physician by her parents for the evaluation of pubic hair development. She has a history of a fracture in each leg and one fracture in her right arm. Her performance at school is good. There is no family history of serious illness. She takes no medications. Vital signs are within normal limits. Genital examination shows coarse, dark hair along the labia. The breast glands are enlarged and the breast bud extends beyond the areolar diameter. There are several hyperpigmented macules with rough, serpiginous borders of different sizes on the lower and upper extremities. The remainder of the examination shows no abnormalities. Which of the following is the most likely diagnosis?

Answer Choices:

A. Osteogenesis imperfecta B. McCune-Albright syndrome C. Neurofibromatosis type I D. Congenital adrenal hyperplasia

Correct Answer: B. McCune-Albright syndrome

Question 164

Clinical Scenario:

A 21-year-old man presents for a pre-employment medical check-up. He has a history of persistent asthma and regularly uses inhaled fluticasone for prophylaxis. For the last week, he has been experiencing increasing symptoms, such as night time cough and wheezing on exertion. Because his albuterol metered-dose inhaler ran out, he has been taking oral albuterol 3 times a day for the last 3 days, which has improved his symptoms. The

physician performs a complete physical examination and orders laboratory tests. Which of the following findings is most likely to be present on his physical examination or laboratory studies?

Answer Choices:

A. Pulse rate is 116/min B. Myoclonus C. Serum potassium is 5.5 mEq/L (5.5 mmol/L) D. Serum magnesium is 2.4 mEq/L (1.2 mmol/L)

Correct Answer: A. Pulse rate is 116/min

Question 165

Clinical Scenario:

A 45-year-old man comes to the emergency department with fever, nonproductive cough, and difficulty breathing. Three years ago, he underwent lung transplantation. A CT scan of the chest shows diffuse bilateral ground-glass opacities. Pathologic examination of a transbronchial lung biopsy specimen shows several large cells containing intranuclear inclusions with a clear halo. Treatment with ganciclovir fails to improve his symptoms. He is subsequently treated successfully with another medication. This drug does not require activation by viral kinases and also has known in-vitro activity against HIV and HBV. The patient was most likely treated with which of the following drugs?

Answer Choices:

A. Foscarnet B. Elvitegravir C. Zanamivir D. Acyclovir

Correct Answer: A. Foscarnet

Question 166

Clinical Scenario:

An 18-year-old man is brought to the emergency department due to the confusion that started earlier in the day. His parents report that the patient had recovered from vomiting and diarrhea 3 days ago without medical intervention. They mention that although nausea and vomiting have resolved, the patient continued to have diffuse abdominal pain and decreased appetite. Past medical history is unremarkable, except for a recent weight loss and increased thirst. The patient does not use tobacco products or alcohol. He is not sexually active and does not use illicit drugs. He appears lethargic but responds to questions. His mucous membranes appear dry. Temperature is 36.9°C (98.4°F), blood pressure is 105/60 mm Hg, pulse is 110/min, and respiratory rate is 27/min with deep and rapid respiration. There is diffuse abdominal tenderness without guarding, rebound tenderness or rigidity. Which of the following is the next best step in the management of this patient?

Answer Choices:

A. ECG B. Abdominal ultrasound C. CT of the abdomen D. Capillary blood glucose measurement

Correct Answer: D. Capillary blood glucose measurement

Question 167

Clinical Scenario:

A 60-year-old man is brought to the emergency department by police officers because he was seen acting strangely in public. The patient was found talking nonsensically to the characters on cereal boxes in a grocery store. Past medical history is significant for multiple hospitalizations for alcohol-related injuries and alcohol withdrawal seizures. Vital signs are within normal limits. On physical examination, the patient is disheveled and oriented x1. Neurologic examination shows horizontal nystagmus and severe ataxia is also noted, that after interviewing the patient, he forgets the face of persons and the questions asked shortly after he walks out the door. He, however, remembers events from his distant past quite well. Which of the following is the most likely diagnosis in this patient?

Answer Choices:

A. Delirium B. Delirium tremens C. Korsakoff amnesia D. Dementia

Correct Answer: C. Korsakoff amnesia

Question 168

Clinical Scenario:

An otherwise healthy 13-year-old boy is brought to the physician because of asthma attacks that have been increasing in frequency and severity over the past 4 weeks. He was first diagnosed with asthma 6 months ago. Current medications include high-dose inhaled fluticasone and salmeterol daily, with additional albuterol as needed. He has required several courses of oral corticosteroids. A medication is added to his therapy regimen that results in downregulation of the high-affinity IgE receptor (FcεRI) on mast cells and basophils. Which of the following drugs was most likely added to the patient's medication regimen?

Answer Choices:

A. Zileuton B. Omalizumab C. Theophylline D. Infliximab

Correct Answer: B. Omalizumab

Question 169

Clinical Scenario:

A 60-year-old woman presents to the emergency department due to progressive shortness of breath and a dry cough for the past week. She notes that her symptoms are exacerbated by physical activity and relieved by rest. The woman was diagnosed with chronic kidney disease 2 years ago and was recently started on regular dialysis treatment. Her pulse rate is 105/min, blood pressure is 110/70 mm Hg, respiratory rate is 30/min, and temperature is 37.8°C (100.0°F). On examination of the respiratory system, there is dullness on percussion, decreased vocal tactile fremitus, and decreased breath sounds over the right lung base. The rest of the physical exam is within normal limits. Which of the following is the most likely cause of this patient's symptoms?

Answer Choices:

A. Primary spontaneous pneumothorax (PSP) B. Pleural effusion C. Pulmonary tuberculosis (TB) D. Acute bronchitis

Correct Answer: B. Pleural effusion

Question 170**Clinical Scenario:**

A scientist wants to extract mRNA from a cell line of interest, amplify a specific mRNA, and insert it into a plasmid so that he can transfect it into a cell in order to over-express that protein. Which of the following proteins is required for the first step of amplification of this mRNA?

Answer Choices:

A. Taq DNA polymerase B. Ligase C. Reverse transcriptase D. RNA polymerase

Correct Answer: C. Reverse transcriptase

Question 171**Clinical Scenario:**

A 35-year-old man comes to the physician because of an ulcer on his penis that he first noticed 4 days ago. He is currently sexually active with multiple male partners and uses condoms inconsistently. Genital examination shows a shallow, nontender ulcer with a smooth base and indurated border along the shaft of the penis. There is bilateral inguinal lymphadenopathy. Darkfield microscopy of a sample from the lesion shows gram-negative, spiral-shaped bacteria. A drug that acts by inhibition of which of the following is the most appropriate treatment for this patient?

Answer Choices:

A. Transpeptidase B. Aminoacyl-tRNA binding C. Dihydrofolate reductase D. Dihydropteroate synthase

Correct Answer: A. Transpeptidase

Question 172**Clinical Scenario:**

A 27-year-old man presents to the emergency department with severe dyspnea and sharp chest pain that suddenly started an hour ago after he finished exercising. He has a history of asthma as a child, and he achieves good control of his acute attacks with Ventolin. On examination, his right lung field is hyperresonant along with diminished lung sounds. Chest wall motion during respiration is asymmetrical. His blood pressure is 105/67 mm Hg, respirations are 22/min, pulse is 78/min, and temperature is 36.7°C (98.0°F). The patient is

supported with oxygen, given corticosteroids, and has had analgesic medications via a nebulizer. Considering the likely condition affecting this patient, what is the best step in management?

Answer Choices:

A. Chest X-rays B. ABG C. Tube insertion D. CT scan

Correct Answer: C. Tube insertion

Question 173

Clinical Scenario:

A 43-year-old construction worker presents to the emergency department two hours after sustaining a deep laceration to his left forearm by a piece of soiled and rusted sheet metal. His vital signs are stable, there is no active bleeding, his pain is well controlled, and a hand surgeon has been notified about damage to his forearm tendons. He does not recall receiving any vaccinations in the last 30 years and does not know if he was vaccinated as a child. What is the appropriate post-exposure prophylaxis?

Answer Choices:

A. IV metronidazole only B. Anthrax vaccine C. Tetanus immunoglobulin only D. Tetanus vaccine + immunoglobulin

Correct Answer: D. Tetanus vaccine + immunoglobulin

Question 174

Clinical Scenario:

A 6-year-old girl is brought to the physician because of a 4-day history of irritation and redness in both eyes. Her symptoms initially started in the left eye and progressed to involve both eyes within 24 hours. She presents with profuse tearing and reports that her eyes are sticky and difficult to open in the morning. She was diagnosed with asthma 2 years ago and has been admitted to the hospital for acute exacerbations 3 times since then. Current medications include inhaled beclomethasone, inhaled albuterol, and montelukast. Her temperature is 38.2 °C (100.8°F). Physical examination reveals a tender left preauricular lymph node. There is chemosis and diffuse erythema of the bulbar conjunctiva bilaterally. Slit lamp examination reveals a follicular reaction in both palpebral conjunctivae and diffuse, fine epithelial keratitis of both corneas. Corneal sensation is normal. Which of the following is the most appropriate next step in management?

Answer Choices:

A. Supportive therapy B. Topical prednisolone acetate C. Topical natamycin D. Topical erythromycin

Correct Answer: A. Supportive therapy

Question 175

Clinical Scenario:

A 20-year-old woman presents with nausea, fatigue, and breast tenderness. She is sexually active with two partners and occasionally uses condoms during intercourse. A β -hCG urinary test is positive. A transvaginal ultrasound reveals an 8-week fetus in the uterine cavity. The patient is distressed by this news and requests an immediate abortion. Which of the following is the most appropriate step in management?

Answer Choices:

A. Ask the patient to reconsider and refer her to a social worker. B. Ask the patient to obtain consent from legal guardians. C. Explain the risk and potential harmful effects of the procedure. D. Ask the patient to obtain consent from the baby's father.

Correct Answer: C. Explain the risk and potential harmful effects of the procedure.

Question 176

Clinical Scenario:

A 62-year-old man is brought to the emergency department by his wife for high blood pressure readings at home. He is asymptomatic. He has a history of hypertension and hyperlipidemia for which he takes atenolol and atorvastatin, however, his wife reports that he recently ran out of atenolol and has not been able to refill it due to lack of health insurance. His temperature is 36.8°C (98.2°F), the pulse 65/min, the respiratory rate 22/min, and the blood pressure 201/139 mm Hg. He has no papilledema on fundoscopic examination. A CT scan shows no evidence of intracranial hemorrhage or ischemia. Of the following, what is the next best step?

Answer Choices:

A. Start or restart low-dose medication to reduce blood pressure gradually over the next 24–48 hours B. Start or restart low-dose medication to reduce blood pressure aggressively over the next 24–48 hours C. Admit him to the ICU and start intravenous medication to reduce blood pressure by 10% in the first hour D. Admit him to the ICU and start intravenous medication to reduce blood pressure by 25% in the first 4 hours

Correct Answer: A. Start or restart low-dose medication to reduce blood pressure gradually over the next 24–48 hours

Question 177

Clinical Scenario:

A 22-year-old college student comes to the physician because of depressed mood and fatigue for the past 5 weeks. He has been feeling sad and unmotivated to attend his college classes. He finds it particularly difficult to get out of bed in the morning. He has difficulty concentrating during lectures and often feels that he is less intelligent compared to his classmates. In elementary school, he was diagnosed with attention deficit hyperactivity disorder and treated with methylphenidate; he stopped taking this medication 4 years ago because

his symptoms had improved during high school. He has smoked two packs of cigarettes daily for 8 years; he feels guilty that he has been unable to quit despite numerous attempts. During his last attempt 3 weeks ago, he experienced increased appetite and subsequently gained 3 kg (6 lb 10 oz) in a week. Mental status examination shows psychomotor retardation and restricted affect. There is no evidence of suicidal ideation. Which of the following is the most appropriate pharmacotherapy?

Answer Choices:

A. Amitriptyline B. Bupropion C. Fluoxetine D. Valproic acid

Correct Answer: B. Bupropion

Question 178

Clinical Scenario:

A 28-year-old man presents with fever, chills, and malaise which began 5 days ago. He also mentions that the back of his right upper arm feels itchy. He says he works as a forest guide and recently came back from a forest expedition. Upon asking, he reports that the forest where he works is infested with ticks. His temperature is 38.3°C (100.9°F), the pulse is 87/min, the respiratory rate is 15/min, and the blood pressure is 122/90 mm Hg. On physical examination, there is a rash present on the posterior aspect of his upper right arm which is shown in the image. Which of the following medications is the best course of treatment for this patient?

Answer Choices:

A. Azithromycin B. Doxycycline C. Fluconazole D. Trimethoprim-sulfamethoxazole

Correct Answer: B. Doxycycline

Question 179

Clinical Scenario:

An investigator is studying the interaction between epithelial cells and calcium ion concentration. When the calcium ion concentration available to a sample of epithelial tissue is decreased, an increased gap between adjacent epithelial cells is seen on electron microscopy. This observed decrease in cell adhesion is most likely due to an effect on which of the following proteins?

Answer Choices:

A. Claudin B. Cadherin C. Actin D. Integrin

Correct Answer: B. Cadherin

Question 180

Clinical Scenario:

A 35-year-old woman presents to the emergency department with swelling of her face and abdominal pain. She states she was outside doing yard work when her symptoms began. The patient has a past medical history of recently diagnosed diabetes and hypertension. Her current medications include lisinopril, metformin, and glipizide. Her temperature is 99.5°F (37.5°C), blood pressure is 149/95 mmHg, pulse is 90/min, respirations are 15/min, and oxygen saturation is 99% on room air. On physical exam, the patient's cardiac and pulmonary exam are within normal limits. Dermatologic exam reveals edema of her hands, lips, and eyelids. There is mild laryngeal edema; however, the patient is speaking clearly and maintaining her airway. Which of the following is appropriate long-term management of this patient?

Answer Choices:

A. Fresh frozen plasma B. Ecallantide C. Danazol D. Discontinue metformin

Correct Answer: C. Danazol

Question 181

Clinical Scenario:

A previously healthy 21-year-old college student is brought to the emergency department because of a 10-hour history of increasing headache, stiff neck, and sensitivity to light. He returned from a mission trip to Haiti 3 weeks ago where he worked in a rural health clinic. He appears lethargic. He is oriented to person, place, and time. His temperature is 39°C (102°F), pulse is 115/min, respirations are 20/min, and blood pressure is 100/70 mm Hg. Examination shows equal and reactive pupils. There are scattered petechiae over the trunk and lower extremities. Range of motion of the neck is decreased due to pain. Neurologic examination shows no focal findings. Blood cultures are obtained and a lumbar puncture is performed. Cerebrospinal fluid (CSF) analysis shows neutrophilic pleocytosis and decreased glucose concentration. Which of the following is most likely to have prevented this patient's condition?

Answer Choices:

A. Fluconazole therapy B. Inactivated whole-cell vaccine C. Toxoid vaccine D. Polysaccharide conjugate vaccine

Correct Answer: D. Polysaccharide conjugate vaccine

Question 182

Clinical Scenario:

A 32-year-old woman comes to the physician because of pain and stiffness in both of her hands for the past 3 weeks. The pain is most severe early in the day and does not respond to ibuprofen. She has no history of serious illness and takes no medications. Vital signs are within normal limits. Examination shows swelling and tenderness of the wrists and metacarpophalangeal joints bilaterally. Range of motion is decreased due to pain. There are subcutaneous, nontender, firm, mobile nodules on the extensor surface of the forearm. Which of the following is the most appropriate pharmacotherapy for this patient's current symptoms?

Answer Choices:

A. Methotrexate B. Colchicine C. Sulfasalazine D. Prednisone

Correct Answer: D. Prednisone

Question 183**Clinical Scenario:**

A 44-year-old woman presents with increased thirst and frequent urination that started 6 months ago and have progressively worsened. Recently, she also notes occasional edema of the face. She has no significant past medical history or current medications. The patient is afebrile and the rest of the vital signs include: blood pressure is 120/80 mm Hg, heart rate is 61/min, respiratory rate is 14/min, and temperature is 36.6°C (97.8°F). The BMI is 35.2 kg/m². On physical exam, there is 2+ pitting edema of the lower extremities and 1+ edema in the face. There is generalized increased deposition of adipose tissue present that is worse in the posterior neck, upper back, and shoulders. There is hyperpigmentation of the axilla and inguinal areas. The laboratory tests show the following findings: Blood Erythrocyte count 4.1 million/mm³ Hgb 12.9 mg/dL Leukocyte count 7,200/mm³ Platelet count 167,000/mm³ Fasting blood glucose 141 mg/dL (7.8 mmol/L) Creatinine 1.23 mg/dL (108.7 µmol/L) Urea nitrogen 19 mg/dL (6.78 mmol/L) Urine dipstick Glucose +++ Protein ++ Bacteria Negative The 24-hour urine protein is 0.36 g. Which of the following medications is the best treatment for this patient's condition?

Answer Choices:

A. Enalapril B. Insulin C. Furosemide D. Mannitol

Correct Answer: A. Enalapril

Question 184**Clinical Scenario:**

A 13-month-old girl is brought to the pediatric clinic by her mother due to progressive abdominal distension, poor feeding, and failure to thrive. The perinatal history was uneventful. The family emigrated from Sudan 8 years ago. The vital signs include: temperature 36.8°C (98.2°F), blood pressure 100/55 mm Hg, and pulse 99/min. The physical examination shows conjunctival pallor, hepatosplenomegaly, and parietal and frontal bossing of the skull. The laboratory test results are as follows: Hemoglobin 8.7 g/dL Mean corpuscular volume 62 µm³ Red cell distribution width 12.2% (normal value is 11.5–14.5%) Reticulocyte count 2.1 % Leukocyte count 10,200/mm³ Platelet count 392,000/mm³ The peripheral blood smear shows microcytic red cells, target cells, and many nucleated red cells. Which of the following is the most likely diagnosis?

Answer Choices:

A. Alpha-thalassemia major B. Glucose-6-phosphate dehydrogenase deficiency C. Sickle cell disease D. Beta-thalassemia major

Correct Answer: D. Beta-thalassemia major

Question 185

Clinical Scenario:

A 42-year-old woman comes to the physician because of urinary leakage over the last year. She reports involuntarily losing small amounts of urine after experiencing a sudden need to void. She has difficulty making it to the bathroom in time, and only feels comfortable going out into public if she has documented the location of all nearby restrooms. She also has begun to wake up at night to urinate. These symptoms have persisted despite 6 months of bladder training and weight loss and reducing soda and coffee intake. Physical examination shows no abnormalities. The most appropriate pharmacotherapy for this patient is a drug that has which of the following mechanisms of action?

Answer Choices:

A. Antagonism of muscarinic M3 receptors B. Antagonism of beta-3 adrenergic receptors C. Agonism of beta-2 adrenergic receptors D. Agonism of muscarinic M2 receptors

Correct Answer: A. Antagonism of muscarinic M3 receptors

Question 186

Clinical Scenario:

A 47-year-old woman comes to the physician because of a 3-day history of fever, fatigue, loss of appetite, cough, and chest pain. Physical examination shows diffuse inspiratory crackles over the left lung field. An x-ray of the chest shows hilar lymphadenopathy and well-defined nodules with central calcifications. Urine studies show the presence of a polysaccharide antigen. A biopsy specimen of the lung shows cells with basophilic, crescent-shaped nuclei and pericellular halos located within macrophages. This patient's history is most likely to show which of the following?

Answer Choices:

A. Treatment with inhaled glucocorticoids B. Recent trip to Brazil C. Previous mycobacterial infection D. Exposure to bat droppings

Correct Answer: D. Exposure to bat droppings

Question 187

Clinical Scenario:

A 38-year-old male presents for counseling by a psychologist mandated by the court. The patient explains that he does not mean to hit his wife when they are arguing, but something just comes over him that he cannot control. Upon further discussion, the patient reveals that his father was incarcerated several times for physically abusing his mother. Which of the following best describes the behavior seen in this patient?

Answer Choices:

A. Acting out B. Identification C. Reaction formation D. Splitting

Correct Answer: B. Identification

Question 188

Clinical Scenario:

A 28-year-old female presents to her primary care provider for headache. The patient reports that every few weeks she has an episode of right-sided, throbbing headache. The episodes began several years ago and are accompanied by nausea and bright spots in her vision. The headache usually subsides if she lies still in a dark, quiet room for several hours. The patient denies any weakness, numbness, or tingling during these episodes. Her past medical history is significant for acne, hypothyroidism, obesity, and endometriosis. Her home medications include levothyroxine, oral contraceptive pills, and topical tretinoin. She has two glasses of wine with dinner several nights a week and has never smoked. She works as a receptionist at a marketing company. On physical exam, the patient has no focal neurologic deficits. A CT of the head is performed and shows no acute abnormalities. Which of the following is the most appropriate treatment for this patient during these episodes?

Answer Choices:

A. Acetazolamide B. High-flow oxygen C. Verapamil D. Sumatriptan

Correct Answer: D. Sumatriptan

Question 189

Clinical Scenario:

A 23-year-old male presents to the emergency room following a gunshot wound to the leg. On arrival his temperature is 99°F (37.2°C), blood pressure is 90/60 mmHg, pulse is 112/min, respirations are 21/min, and pulse oximetry is 99% on room air. Two large bore IVs are placed and he receives crystalloid fluid replacement followed by 2 units of crossmatched packed red blood cells. Immediately following transfusion, his temperature is 102.2°F (39°C), blood pressure is 93/64 mmHg, pulse is 112/min, respirations are 21/min, and pulse oximetry is 99% on room air. There is oozing from his IV sites. You check the records and realize there was a clerical error with the blood bank. What is the mechanism for his current condition?

Answer Choices:

A. Preformed antibodies B. Deposition of immune complexes C. T lymphocyte reaction D. Production of leukotrienes

Correct Answer: A. Preformed antibodies

Question 190

Clinical Scenario:

A 45-year-old woman comes to the emergency department with severe menorrhagia for 3 days. She also reports dizziness. She has hypertension, for which she takes lisinopril. She appears pale. Her temperature is 37.5°C (99.5°F), pulse is 110/min, and blood pressure is 100/60 mmHg. Pulse oximetry shows an oxygen saturation of 98% on room air. Pelvic examination shows vaginal vault with dark maroon blood and clots but no active source of bleeding. Her hemoglobin concentration is 5.9 g/dL. Crystalloid fluids are administered and she is transfused with 4 units of crossmatched packed red blood cells. Two hours later, she has shortness of breath and dull chest pressure. Her temperature is 37.6°C (99.7°F), pulse is 105/min, and blood pressure is 170/90 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 92% on room air. Cardiac examination shows an S3 gallop. Diffuse crackles are heard over the lower lung fields on auscultation. An x-ray of the chest shows bilateral hazy opacities. An ECG shows no abnormalities. Which of the following is the most likely explanation of this patient's symptoms?

Answer Choices:

A. Type 1 hypersensitivity reaction B. Acute pulmonary embolism C. Acute kidney injury D. Transfusion-associated circulatory overload

Correct Answer: D. Transfusion-associated circulatory overload

Question 191

Clinical Scenario:

A 62-year-old retired professor comes to the clinic with the complaints of back pain and increasing fatigue over the last 4 months. For the past week, his back pain seems to have worsened. It radiates to his legs and is burning in nature, 6/10 in intensity. There is no associated tingling sensation. He has lost 4.0 kg (8.8 lb) in the past 2 months. There is no history of trauma. He has hypertension which is well controlled with medications. Physical examination is normal. Laboratory studies show normocytic normochromic anemia. Serum calcium is 12.2 mg/dL and Serum total proteins is 8.8 gm/dL. A serum protein electrophoresis shows a monoclonal spike. X-ray of the spine shows osteolytic lesions over L2–L5 and right femur. A bone marrow biopsy reveals plasmacytosis. Which of the following is the most preferred treatment option?

Answer Choices:

A. Bisphosphonates B. Chemotherapy alone C. Renal dialysis D. Chemotherapy and autologous stem cell transplant

Correct Answer: D. Chemotherapy and autologous stem cell transplant

Question 192

Clinical Scenario:

A 21-year-old male college student is brought to the emergency department by the campus police after he was found yelling at a bookshelf in the library. His roommate does not know of any prior episodes similar to this. His vital signs are within normal limits. The patient appears unkempt. On mental status examination, he talks very fast with occasional abrupt interruptions. He is agitated. He is disoriented to time and repeatedly tells the physician, "I hear the sun telling me that I was chosen to save the universe." Urine toxicology screen is negative. Which of the following is the most appropriate pharmacotherapy?

Answer Choices:

A. Haloperidol B. Dexmedetomidine C. Valproic acid D. Ziprasidone

Correct Answer: D. Ziprasidone

Question 193

Clinical Scenario:

A 20-year-old man presents to the emergency department by his father for not sleeping for 2 nights consecutively. His father noticed that the patient has been in an unusual mood. One day ago, the patient disrobed in front of guests after showering. He has also had lengthy conversations with strangers. One month ago, the patient took out a large loan from a bank in order to fund a business idea he has not yet started. He also borrowed his father's credit card to make a spontaneous trip to Switzerland by himself for a few days, where he spent over 30,000 dollars. His father notes that there have been episodes where he would not leave his bed and remained in his room with the lights off. During these episodes, he sleeps for approximately 15 hours. On physical exam, he is talkative, distractable, and demonstrates a flight of ideas. His speech is pressured, difficult to interrupt, and he asks intrusive questions. Which of the following is the best treatment option for this patient?

Answer Choices:

A. Carbamazepine B. Escitalopram C. Lithium D. Observation

Correct Answer: C. Lithium

Question 194

Clinical Scenario:

A 50-year-old man presents to his primary care provider complaining of smelling abnormal odors on several occasions. He says that he smells burnt rubber even though there is nothing burning and no one around him can smell what he does. This symptom has been intermittently bothering him for the past 6 months. Also during this period, he had occasional nosebleeds. He works as a high school teacher. Although his work gets a little stressful around the exam season, he says he is able to cope well. Family history is unremarkable. He does not smoke or drink alcohol and denies the use of any medication. Physical examination reveals unilateral nasal obstruction with some dried blood in the nasal passage. What is the most likely diagnosis?

Answer Choices:

A. Psychomotor epilepsy B. Neuroblastoma C. Hypnagogic hallucination D. Schizophrenia

Correct Answer: B. Neuroblastoma

Question 195

Clinical Scenario:

A 45-year-old Caucasian male presents complaining of inability to open his mouth. Patient history reveals that he recently injured his foot from an exposed floor nail in his house. This patient's symptoms are likely the result of:

Answer Choices:

A. Impaired motor neuron release of ACh B. Impaired motor neuron release of GABA C. Cross-reactivity of bacterial antigens D. Bacterial infiltration of the central nervous system

Correct Answer: B. Impaired motor neuron release of GABA

Question 196

Clinical Scenario:

A 40-year-old man comes to the physician because of a 5-month history of watery diarrhea and episodic crampy abdominal pain. He has no fever, nausea, or vomiting. Over the past 6 months, he has had a 1.8-kg (4-lb) weight loss, despite experiencing no decrease in appetite. His wife has noticed that sometimes his face and neck become red after meals or when he is in distress. A year ago, he was diagnosed with asthma. He has hypertension. Current medications include an albuterol inhaler and enalapril. He drinks one beer daily. His temperature is 36.7°C (98°F), pulse is 85/min, and blood pressure is 130/85 mm Hg. The lungs are clear to auscultation. A grade 2/6 systolic murmur is heard best at the left sternal border and fourth intercostal space. The abdomen is soft, and there is mild tenderness to palpation with no guarding or rebound. The remainder of the physical examination shows no abnormalities. A complete blood count is within the reference range. Without treatment, this patient is at greatest risk for which of the following conditions?

Answer Choices:

A. Asphyxia B. Achlorhydria C. Dementia D. Intestinal fistula

Correct Answer: C. Dementia

Question 197

Clinical Scenario:

A 17-year-old boy presents to the office with allergic rhinitis. He reports symptoms of sneezing, nasal congestion, itching, and postnasal drainage every September at the start of the school year. He has a family

history of childhood asthma and eczema. He has not tried any medications for his allergies. Which of the following medications is the most appropriate next step to manage the patient's symptoms?

Answer Choices:

A. Intranasal antihistamines B. Intranasal cromolyn sodium C. Intranasal decongestants D. Intranasal corticosteroids

Correct Answer: D. Intranasal corticosteroids

Question 198

Clinical Scenario:

A 63-year-old woman presents with dyspnea on exertion. She reports that she used to work in her garden without any symptoms, but recently she started to note dyspnea and fatigue after working for 20–30 minutes. She has type 2 diabetes mellitus diagnosed 2 years ago but she does not take any medications preferring natural remedies. She also has arterial hypertension and takes torsemide 20 mg daily. The weight is 88 kg and the height is 164 cm. The vital signs include: blood pressure is 140/85 mm Hg, heart rate is 90/min, respiratory rate is 14/min, and the temperature is 36.6°C (97.9°F). Physical examination is remarkable for increased adiposity, pitting pedal edema, and present S3. Echocardiography shows a left ventricular ejection fraction of 51%. The combination of which of the following medications would be a proper addition to the patient's therapy?

Answer Choices:

A. Metoprolol and indapamide B. Enalapril and bisoprolol C. Indapamide and amlodipine D. Valsartan and spironolactone

Correct Answer: B. Enalapril and bisoprolol

Question 199

Clinical Scenario:

A 32-year-old man is brought to the emergency department with fever, dyspnea, and impaired consciousness. His wife reports that he has also had an episode of dark urine today. Two weeks ago, he returned from a trip to the Republic of Congo. His temperature is 39.4°C (103°F), pulse is 114/min, and blood pressure is 82/51 mm Hg. Physical examination shows scleral icterus. Decreased breath sounds and expiratory crackles are heard on auscultation of the lungs bilaterally. His hemoglobin concentration is 6.3 g/dL. A blood smear shows red blood cells with normal morphology and ring-shaped inclusions. Further laboratory testing shows normal rates of NADPH production. Which of the following is the most appropriate pharmacotherapy for this patient?

Answer Choices:

A. Proguanil B. Chloroquine C. Dapsone D. Artesunate

Correct Answer: D. Artesunate

Question 200

Clinical Scenario:

A 21-year-old woman with type 1 diabetes mellitus suddenly develops tremors, cold sweats, and confusion while on a backpacking trip with friends. She is only oriented to person and is unable to follow commands. Her fingerstick blood glucose concentration is 28 mg/dL. Her friend administers an intramuscular injection with a substance that reverses her symptoms. Which of the following is the most likely mechanism of action of this drug?

Answer Choices:

A. Activation of glucokinase B. Inhibition of glucose-6-phosphatase C. Inhibition of α -glucosidase D. Activation of adenylyl cyclase

Correct Answer: D. Activation of adenylyl cyclase

Question 201

Clinical Scenario:

A 3-year-old child is brought to the emergency department by his parents. The child presents with significant rapid breathing and appears unwell. On examination, his liver size is 1.5 times larger than children of his age, and he has mild pitting edema in his legs. This child is also in the lower weight-age and height-age percentiles. On auscultation, mild rales were noted and a fixed split S2 was heard on inspiration. There is no family history of congenital disorders or metabolic syndromes. Which of the following is the likely diagnosis?

Answer Choices:

A. Liver failure B. Atrial septal defect C. Patent foramen ovale D. Endocardial cushion syndrome

Correct Answer: B. Atrial septal defect

Question 202

Clinical Scenario:

A 13-year-old boy presents to his pediatrician with a 1-day history of frothy brown urine. He says that he believes he had strep throat some weeks ago, but he was not treated with antibiotics as his parents were worried about him experiencing harmful side effects. His blood pressure is 148/96 mm Hg, heart rate is 84/min, and respiratory rate is 15/min. Laboratory analysis is notable for elevated serum creatinine, hematuria with RBC casts, and elevated urine protein without frank proteinuria. His antistreptolysin O titer is elevated, and he is subsequently diagnosed with post-streptococcal glomerulonephritis (PSGN). His mother is distraught regarding the diagnosis and is wondering if this could have been prevented if he had received antibiotics. Which of the following is the most appropriate response?

Answer Choices:

A. Antibiotic therapy can prevent the development of PSGN. B. Once a patient is infected with a nephritogenic strain of group A streptococcus, the development of PSGN cannot be prevented. C. Antibiotic therapy only prevents PSGN in immunosuppressed patients. D. Antibiotic therapy decreases the severity of PSGN.

Correct Answer: B. Once a patient is infected with a nephritogenic strain of group A streptococcus, the development of PSGN cannot be prevented.

Question 203

Clinical Scenario:

A 52-year-old woman comes to the physician because of a 6-month history of generalized fatigue, low-grade fever, and a 10-kg (22-lb) weight loss. Physical examination shows generalized pallor and splenomegaly. Her hemoglobin concentration is 7.5 g/dL and leukocyte count is 41,800/mm³. Leukocyte alkaline phosphatase activity is low. Peripheral blood smear shows basophilia with myelocytes and metamyelocytes. Bone marrow biopsy shows cellular hyperplasia with proliferation of immature granulocytic cells. Which of the following mechanisms is most likely responsible for this patient's condition?

Answer Choices:

A. Cytokine-independent activation of the JAK-STAT pathway B. Loss of function of the APC gene C. Altered expression of the retinoic acid receptor gene D. Unregulated expression of the ABL1 gene

Correct Answer: D. Unregulated expression of the ABL1 gene

Question 204

Clinical Scenario:

A 62-year-old man with a past medical history notable for α -thalassemia now presents for urgent care with complaints of increased thirst and urinary frequency. Physical examination is grossly unremarkable, although there is a bronze discoloration of his skin. His vital signs include: temperature 36.7°C (98.0°F), blood pressure 126/74 mm Hg, heart rate 74/min, and respiratory rate 14/min. Laboratory analysis reveals fasting blood glucose of 192 mg/dL and subsequently, HbA1c of 8.7. Given the following options, what is the definitive treatment for the patient's underlying disease?

Answer Choices:

A. Metformin B. Basal and bolus insulin C. Recurrent phlebotomy D. Deferoxamine

Correct Answer: C. Recurrent phlebotomy

Question 205

Clinical Scenario:

A group of gastroenterologists is concerned about low colonoscopy screening rates. They decide to implement a free patient navigation program to assist local residents and encourage them to obtain colonoscopies in accordance with U.S. Preventive Services Task Force (USPSTF) guidelines. Local residents were recruited at community centers. Participants attended monthly meetings with patient navigators and were regularly reminded that their adherence to screening guidelines was being evaluated. Colonoscopy screening rates were assessed via chart review, which showed that 90% of participants adhered to screening guidelines. Data collected via chart review for local residents recruited at community centers who did not participate in the free patient navigation system found that 34% of that population adhered to USPSTF guidelines. Which of the following has most likely contributed to the observed disparity in colonoscopy screening rates?

Answer Choices:

A. Confirmation bias B. Hawthorne effect C. Sampling bias D. Recall bias "

Correct Answer: B. Hawthorne effect

Question 206

Clinical Scenario:

An American pediatrician travels to Bangladesh on a medical mission. While working in the local hospital's emergency room, she sees a 2-week-old boy who was brought in by his mother with muscle spasms and difficulty sucking. The mother gave birth at home at 38 weeks gestation and was attended to by her older sister who has no training in midwifery. The mother had no prenatal care. She has no past medical history and takes no medications. The family lives on a small fishing vessel on a major river, which also serves as their fresh water supply. The boy's temperature is 99°F (37.2°C), blood pressure is 100/60 mmHg, pulse is 130/min, and respirations are 22/min. On exam, the boy's arms are flexed at the elbow, his knees are extended, and his neck and spine are hyperextended. Tone is increased in the bilateral upper and lower extremities. He demonstrates sustained facial muscle spasms throughout the examination. The umbilical stump is foul-smelling. Cultures are taken, and the appropriate treatment is started. This patient's condition is most likely caused by a toxin with which of the following functions?

Answer Choices:

A. Binding to MHC II and the T cell receptor simultaneously B. Blocking release of acetylcholine C. Blocking release of GABA and glycine D. Blocking voltage-gated sodium channel opening

Correct Answer: C. Blocking release of GABA and glycine

Review Summary

Category: Treatment/Management **Questions Reviewed:** ____ / 206

Overall Assessment: - Correctly categorized: _ questions - Miscategorized: questions - Ambiguous: __ questions

General Comments:

Reviewer Name: _____

Date: _____

Credentials: _____